

OPTIMAL LINEAR FILTERING IN MEAN SQUARE SENSE: FIR AND IIR WIENER FILTERS



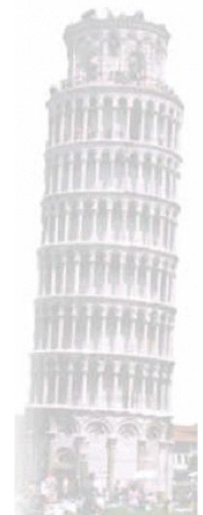
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- The goal is to develop LTI filters able to process in an optimal way a class of statistically equivalent signals.
- As the LMMSE estimate and the optimal linear prediction, the optimal linear filtering is based on the **orthogonality principle**.
- As we have seen, we have 4 different types of **Wiener Filter**, depending on the choice of θ (the random vector to be estimated) and on the data to be used for its estimate: (1) **Filtering**, (2) **Smoothing**, (3) **Prediction**, and (4) **Interpolation**.
- The “pure” linear prediction is a particular case of optimal linear filtering in mean square sense.
- **Filtering problem**: design the optimal Wiener filter of FIR (Finite Impulse Response) or IIR (Infinite Impulse Response) type, causal e not-causal, for the estimate of a random signal in the presence of additive noise (not necessarily white or Gaussian).

- We want to estimate the value at the discrete-time n of a WSS process $S[n]$, by using the present and past (in the case of *smoothing* also the future) observed samples of a noisy version of the signal process.
- **Assumption 1:** the noise is **additive**, i.e. the observed process is:

$$X[n] = S[n] + W[n]$$

Processo sta zionario

- Signal and noise processes are uncorrelated:

$$R_{SW}[m] = E\{S[n]W[n+m]\} = 0, \quad \forall m$$

- **Assumption 2:** the signal is a zero mean process with known power spectral density (PSD).
- **Assumption 3:** the noise is a zero mean process with known power spectral density (PSD).

- Since signal and noise processes are uncorrelated, the ACF and the PSD of the observed process $X[n]=S[n]+W[n]$ are given by:

$$R_X[m] = R_S[m] + R_W[m] \Rightarrow S_X(e^{j2\pi f}) = S_S(e^{j2\pi f}) + S_W(e^{j2\pi f})$$

- The cross-correlation function between the signal of interest $S[n]$ (to be estimated) and the observed signal $X[n]$ is:

più gli spettri sono diversi meglio è: poiché questo filtro si basa sugli spettri

$$R_{SX}[m] = E\{S[n]X[n+m]\} = E\{S[n](S[n+m] + W[n+m])\}$$

↳ Segnale bianco + Rumore Bianco = White

$$= R_S[m] + R_{SW}[m] = R_S[m]$$

- The impulse response $h[n]$ of the FIR and IIR filters (causal or not causal), can be obtained by applying the **Orthogonality Principle**.

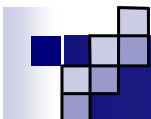
$$\hat{S}[n] = \sum_{k=0}^P h[k] X[n-k], \quad \text{FIR causal Wiener Filter}$$

$$\hat{S}[n] = \sum_{k=0}^{+\infty} h[k] X[n-k], \quad \text{IIR causal Wiener Filter}$$

Versione TV
del filtro wiener in modo
smoothing. Versione sequenziale
(stazionaria)

$$\hat{S}[n] = \sum_{k=-\infty}^{+\infty} h[k] X[n-k], \quad \text{IIR non causal Wiener Filter}$$

- All the filters are linear and time-invariant (LTI), the first two are causal, while the third one is not causal (the LTI **Wiener *smoothing* filter**).



- The **FIR causal Wiener filter** of order P :

$$\hat{S}[n] = \sum_{k=0}^P h[k]X[n-k], \quad \text{FIR causal Wiener filter}$$

- The **estimation error** $\varepsilon[n]$ is given by:

$$\varepsilon[n] = S[n] - \hat{S}[n] = S[n] - \sum_{k=0}^P h[k]X[n-k]$$

- The coefficients of the FIR Wiener filter $h[n]$, $n=0,1,2,\dots,P$, should be obtained by minimizing the mean squared error (MSE), or equivalently, by applying the **Orthogonality Principle**:

$$\min_{\{h[n]\}} \sigma_{\varepsilon}^2 = E\{\varepsilon^2[n]\} \Rightarrow E\{\varepsilon[n]X[n-m]\} = 0, \quad m = 0, 1, 2, \dots, P$$

$$\begin{aligned} E\{\varepsilon[n]X[n-m]\} &= E\left\{\left(S[n] - \hat{S}[n]\right)X[n-m]\right\} \\ &= E\left\{\left(S[n] - \sum_{k=0}^P h[k]X[n-k]\right)X[n-m]\right\} \\ &= E\{S[n]X[n-m]\} - \sum_{k=0}^P h[k]E\{X[n-k]X[n-m]\} \\ &= R_{SX}[-m] - \sum_{k=0}^P h[k]R_X[m-k] = 0, \quad 0 \leq m \leq P \end{aligned}$$

■ The discrete-time Wiener-Hopf equation:

$$\sum_{k=0}^P h[k]R_X[m-k] = R_{SX}[-m], \quad 0 \leq m \leq P$$

- If signal and noise processes are uncorrelated:

$$R_{SX}[-m] = R_S[-m] = R_S[m]$$

- The discrete-time **Wiener-Hopf equation** becomes:

$$\sum_{k=0}^P h[k] R_X[m-k] = R_S[m], \quad 0 \leq m \leq P$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{\varepsilon[n]S[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{S^2[n]\} - \sum_{k=0}^P h[k] E\{S[n]X[n-k]\} \\ &= R_S[0] - \sum_{k=0}^P h[k] R_{SX}[-k] = R_S[0] - \sum_{k=0}^P h[k] R_S[k] \end{aligned}$$

- The linear system of $P+1$ equations can be rewritten in a matrix form:

$$\begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] & \cdots & R_X[-P] \\ R_X[1] & R_X[0] & \ddots & \ddots & R_X[-P+1] \\ R_X[2] & R_X[1] & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & R_X[-1] \\ R_X[P] & R_X[P-1] & \cdots & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \\ \vdots \\ h[P] \end{bmatrix} = \begin{bmatrix} R_S[0] \\ R_S[1] \\ R_S[2] \\ \vdots \\ R_S[P] \end{bmatrix}$$

- By defining, as usual:

$$\mathbf{C}_X \triangleq \begin{bmatrix} R_X[0] & R_X[-1] & \cdots & R_X[-P] \\ R_X[1] & R_X[0] & \cdots & R_X[-(P-1)] \\ \vdots & \vdots & \ddots & \vdots \\ R_X[P] & R_X[P-1] & \cdots & R_X[0] \end{bmatrix}, \quad \mathbf{c}_S \triangleq \begin{bmatrix} R_S[0] \\ R_S[1] \\ \vdots \\ R_S[P] \end{bmatrix}, \quad \mathbf{h} \triangleq \begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[P] \end{bmatrix}$$

$\mathbf{C}_X \mathbf{h} = \mathbf{c}_S \rightarrow$ Matrice sempre invertibile

- By solving with respect to (w.r.t.) $\mathbf{h}$, e.g. through the **Levinson-Durbin** recursive algorithm, if the covariance matrix $\mathbf{C}_X$ (symmetric and with Toeplitz structure) is full rank, we obtain the impulse response $\mathbf{h}$ of the **FIR causal Wiener filter**:

$$\mathbf{h} = \mathbf{C}_X^{-1} \mathbf{c}_S = (\mathbf{C}_S + \mathbf{C}_W)^{-1} \mathbf{c}_S$$

$$\sigma_\varepsilon^2 = R_S[0] - \sum_{k=0}^P h[k] R_S[k]$$

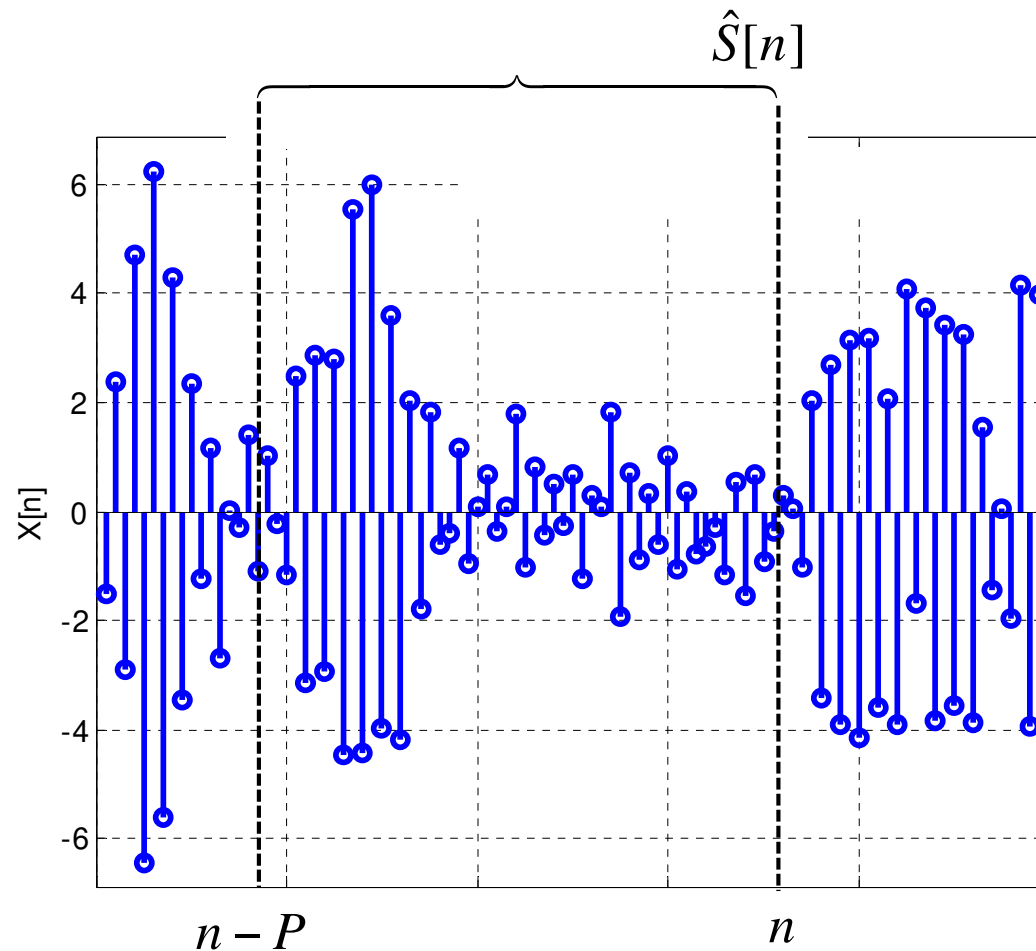
$$= R_S[0] - \mathbf{h}^T \mathbf{c}_S$$

$$= R_S[0] - \mathbf{c}_S^T \mathbf{C}_X^{-1} \mathbf{c}_S$$

$$= R_S[0] - \mathbf{c}_S^T (\mathbf{C}_S + \mathbf{C}_W)^{-1} \mathbf{c}_S$$

↓
banda stretta → più correlazione (meno var.)
→ più alti coeff. → minore MSE

$$\hat{S}[n] = \sum_{k=0}^P h[k] X[n-k], \quad \text{FIR causal Wiener filter}$$



■ **Example:** $S[n]$ is an AR(1) process and $W[n]$ is additive white noise.

We want to find the **FIR Wiener filter** of order $P=2$.

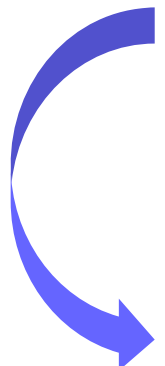
Matrice da inv $P+1$

$$\left\{ \begin{array}{l} X[n] = S[n] + W[n] \\ R_S[m] = \sigma_S^2 \rho^{|m|}, \quad \sigma_S^2 = 2, \quad \rho = 0.8, \quad R_W[m] = \sigma_W^2 \delta[m], \quad \sigma_W^2 = 2 \\ R_X[m] = R_S[m] + R_W[m], \quad R_{SX}[m] = R_S[m], \quad \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2} \end{array} \right.$$

$$\hat{S}[n] = \sum_{k=0}^2 h[k] X[n-k], \quad \text{FIR causal Wiener Filter}$$

$$\mathbf{h} = \mathbf{C}_X^{-1} \mathbf{c}_S = (\mathbf{C}_S + \mathbf{C}_W)^{-1} \mathbf{c}_S$$

- The system of 3 linear equations can be rewritten in matrix form:

$$\begin{bmatrix} R_x[0] & R_x[-1] & R_x[-2] \\ R_x[1] & R_x[0] & R_x[-1] \\ R_x[2] & R_x[1] & R_x[0] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[0] \\ R_s[1] \\ R_s[2] \end{bmatrix}$$


$$2 \cdot \begin{bmatrix} 2 & 0.8 & 0.64 \\ 0.8 & 2 & 0.8 \\ 0.64 & 0.8 & 2 \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 0.8 \\ 0.64 \end{bmatrix}$$

- The solution is given by:

$$\mathbf{h} = \begin{bmatrix} h[0] \\ h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 2 & 0.8 & 0.64 \\ 0.8 & 2 & 0.8 \\ 0.64 & 0.8 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0.8 \\ 0.64 \end{bmatrix} = \begin{bmatrix} 0.3824 \\ 0.2000 \\ 0.1176 \end{bmatrix}$$

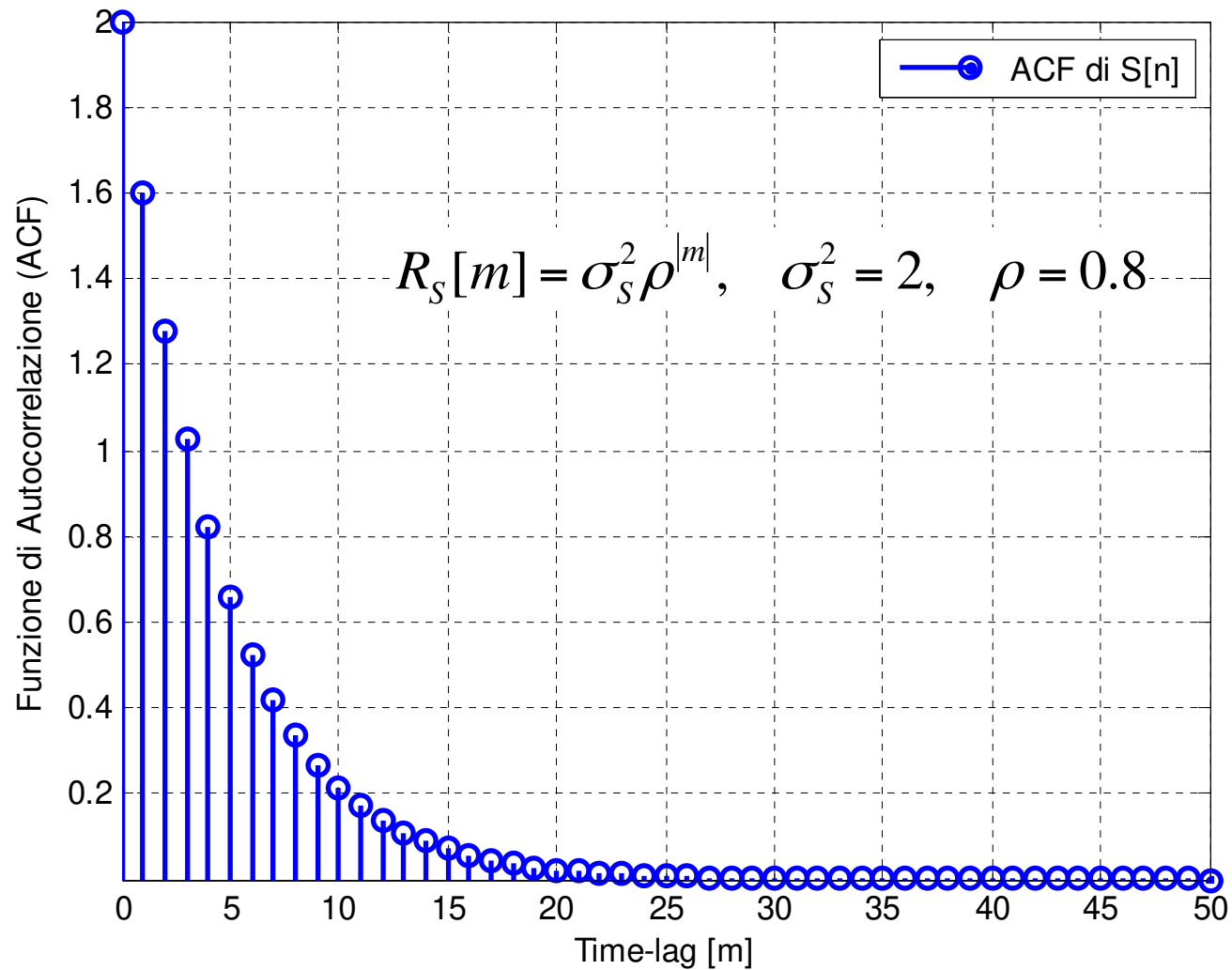
- The signal estimate obtained by using the FIR(2) Wiener filter is:

$$\hat{S}[n] = 0.3824 \cdot X[n] + 0.2 \cdot X[n-1] + 0.1176 \cdot X[n-2]$$

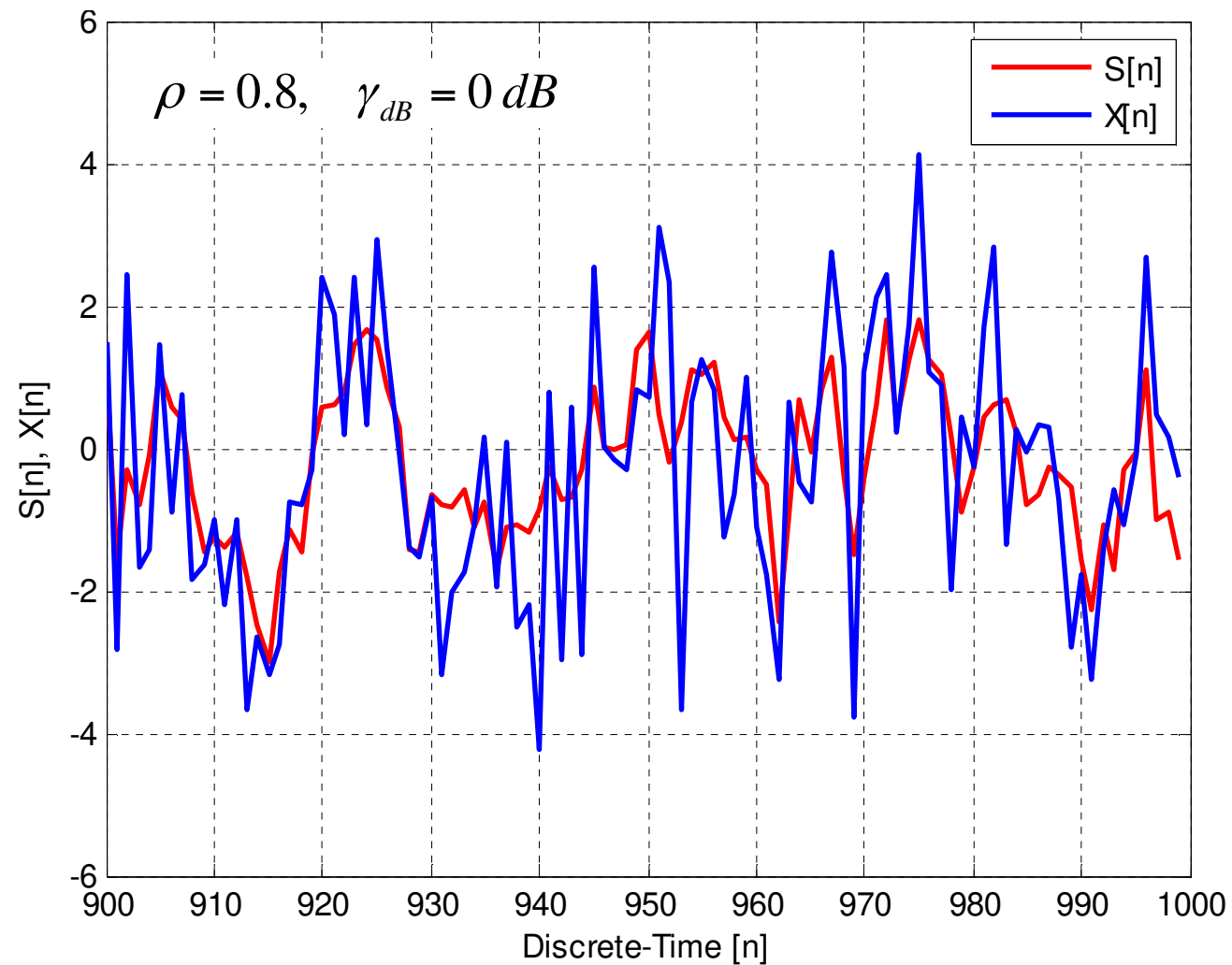
- The MSE of the estimate is:

$$\begin{aligned}\sigma_{\varepsilon}^2 &= R_S[0] - \sum_{k=0}^P h[k] R_S[k] = \sigma_S^2 - \sum_{k=0}^2 h[k] \sigma_S^2 \rho^k \\ &= \sigma_S^2 \left[1 - \sum_{k=0}^2 h[k] \rho^k \right] \\ &= 2[1 - 0.3824 - 0.2 \cdot 0.8 - 0.1176 \cdot 0.64] \\ &= 0.7647\end{aligned}$$

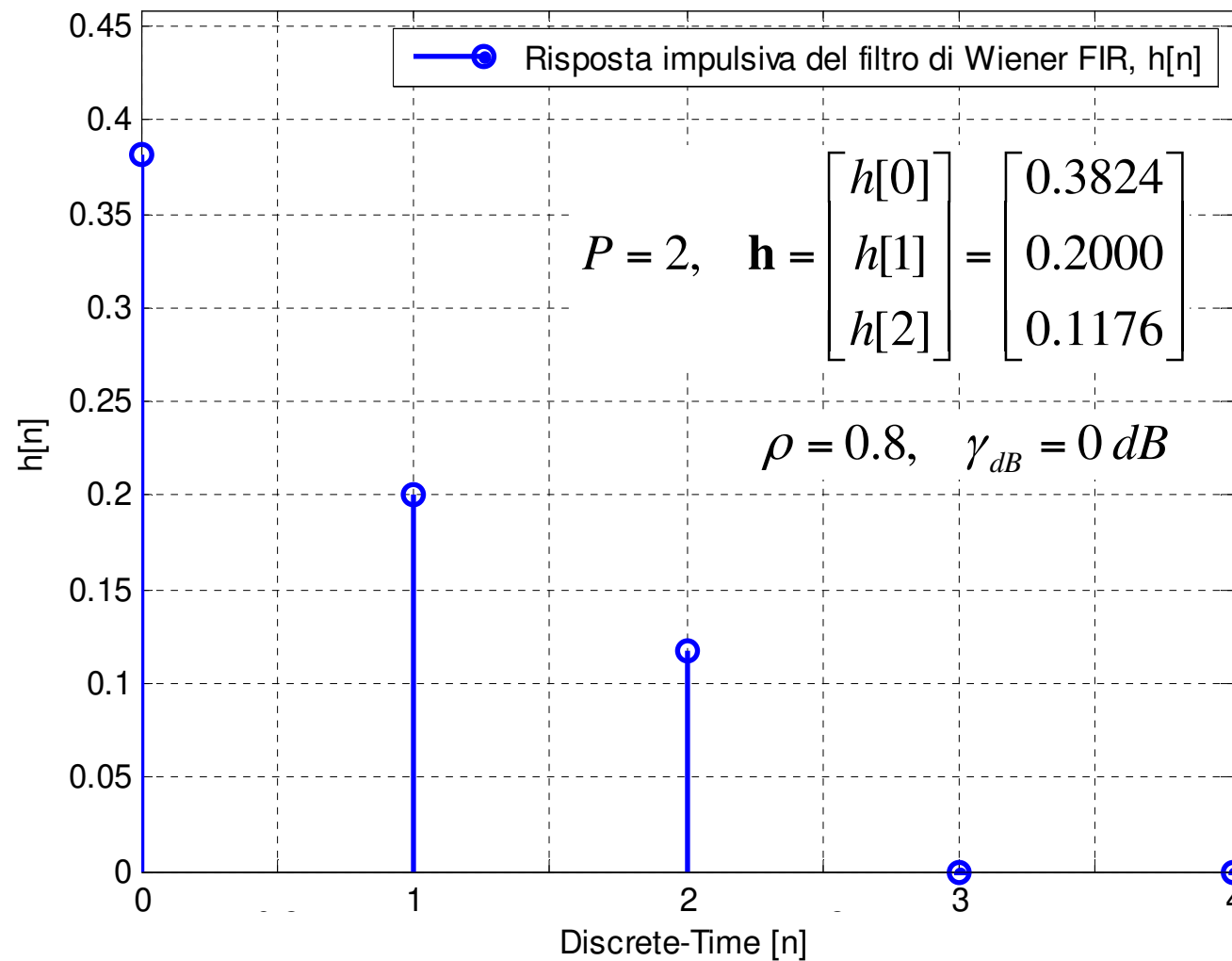
$S[n]$ = Gaussian AR(1) process



$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



Impulse Response (IR) of the FIR(2) Wiener filter



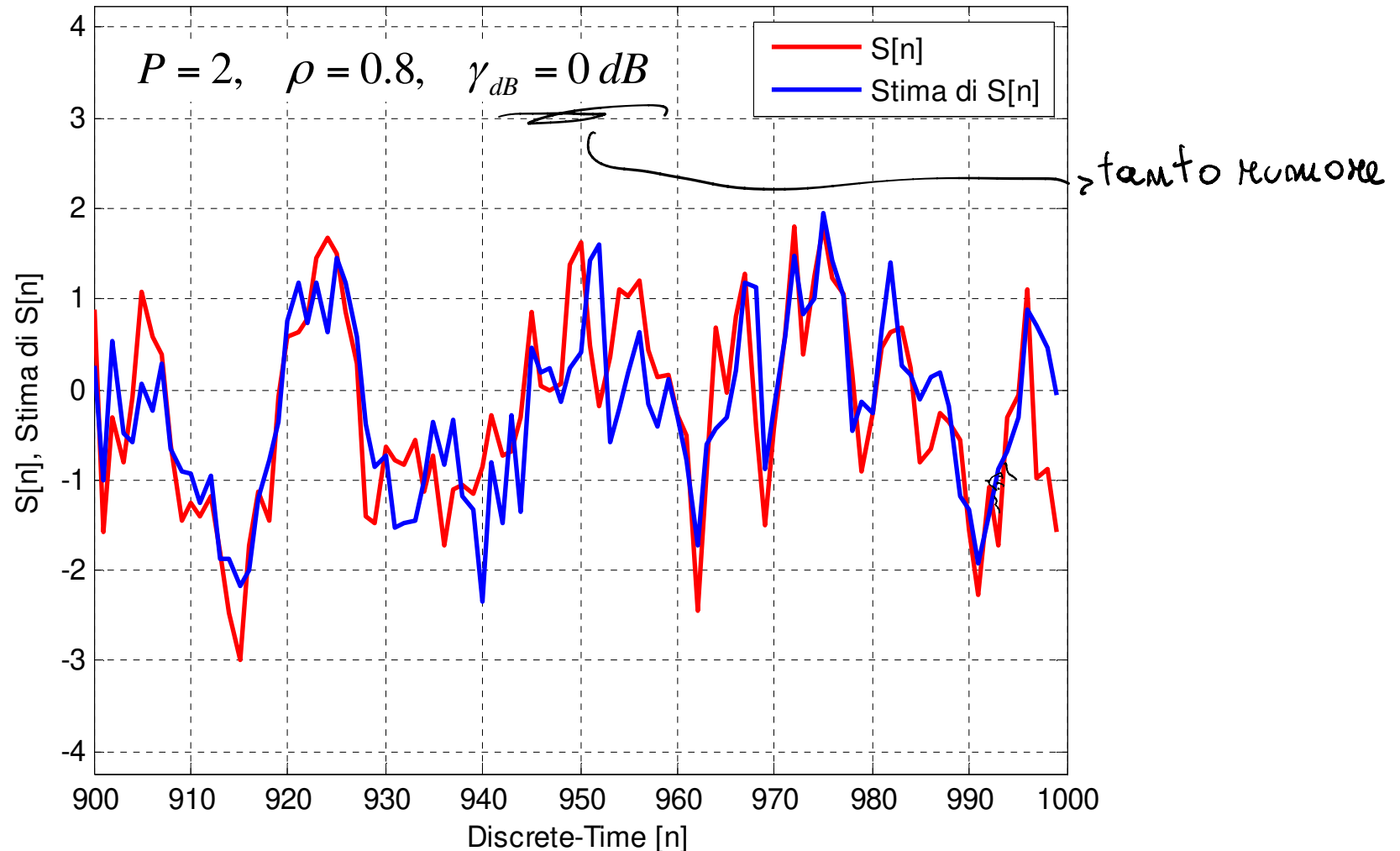
MSE migliora se $\rho \uparrow$ (Rumore bianco), meno banda.

Filtering: FIR Wiener Filter

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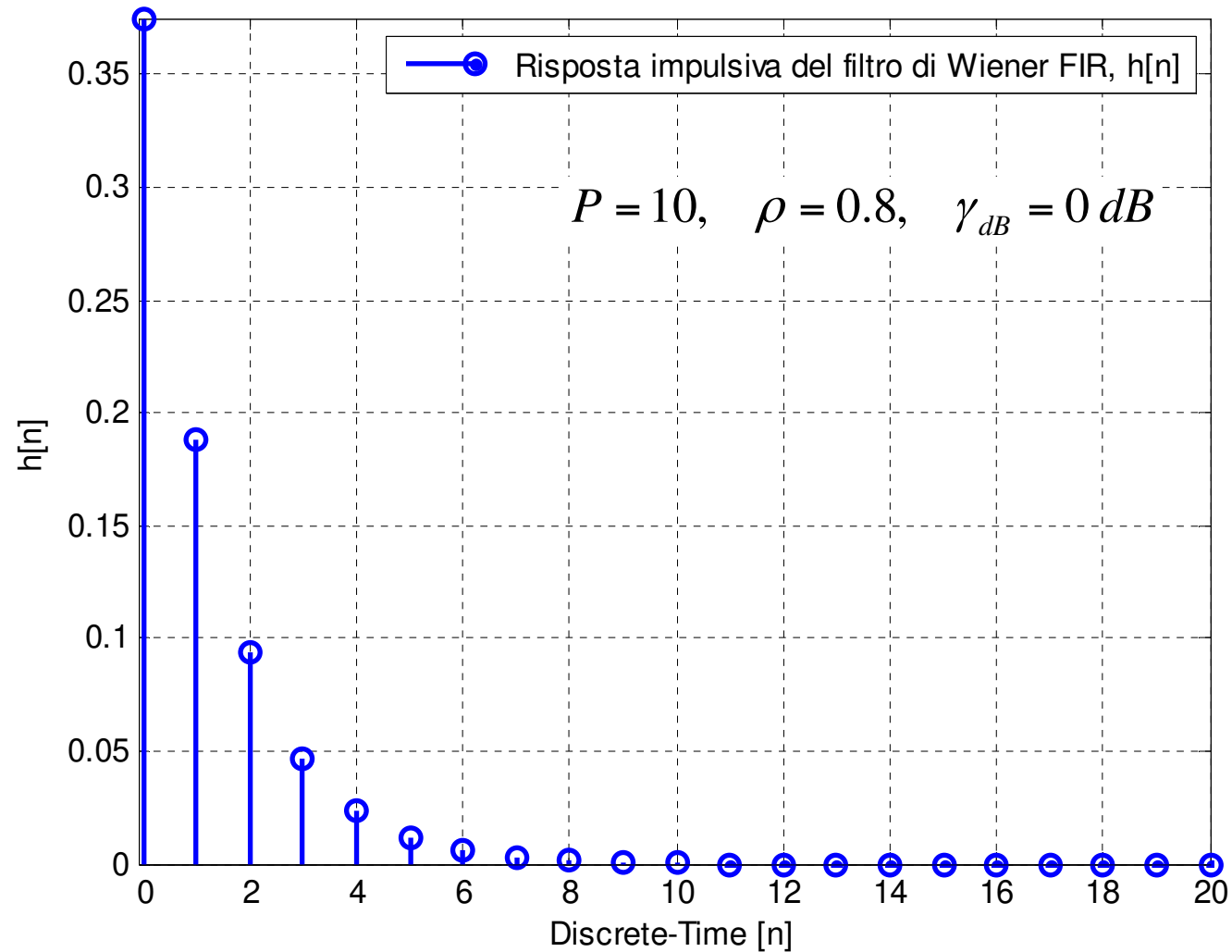
↳ ~~same~~ stesso tipo di quello che studio

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN

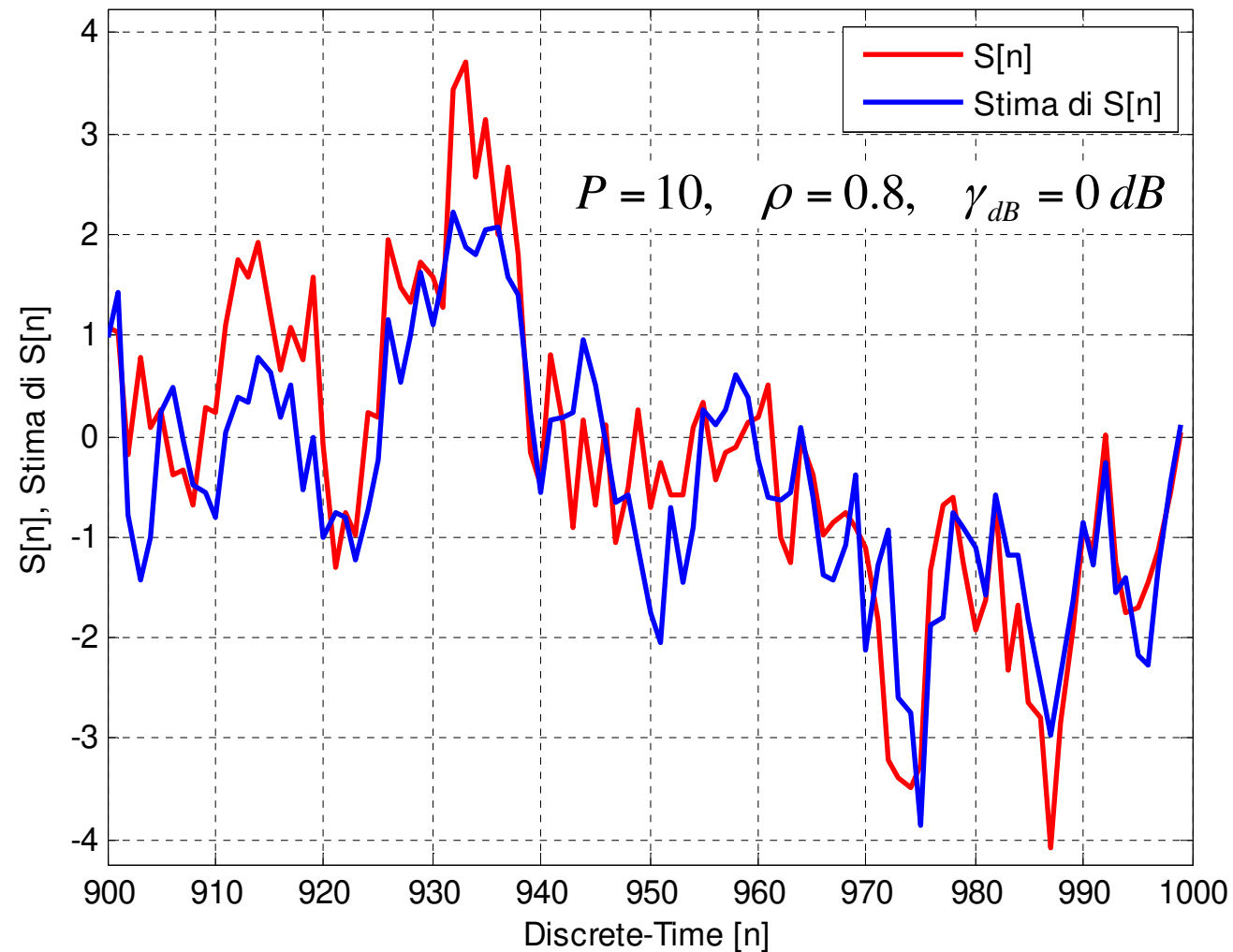


Banda dipende dal rumore, più freq. fogli (AR(1))
più diminuisce var: signal.

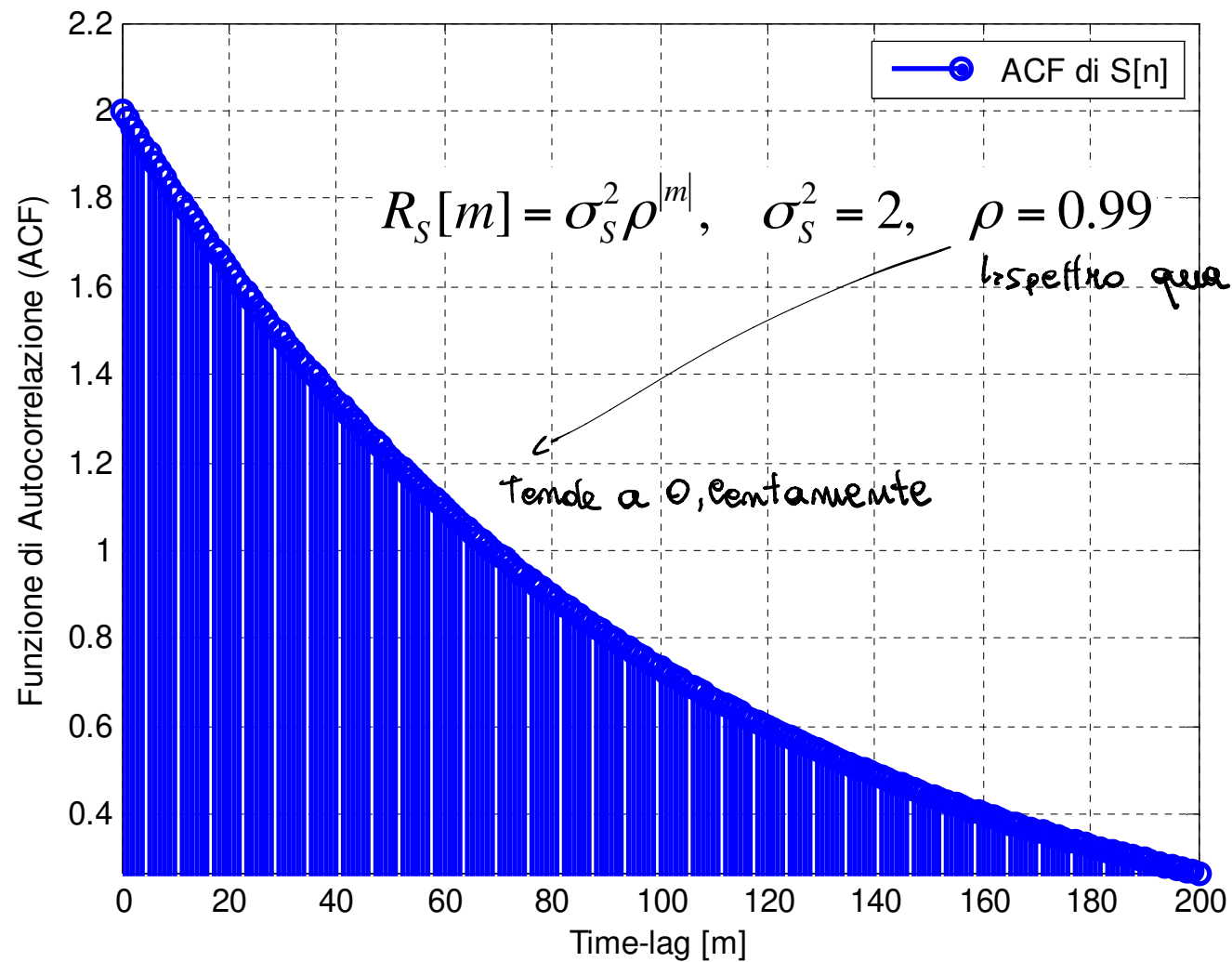
Impulse Response (IR) of the FIR(2) Wiener filter



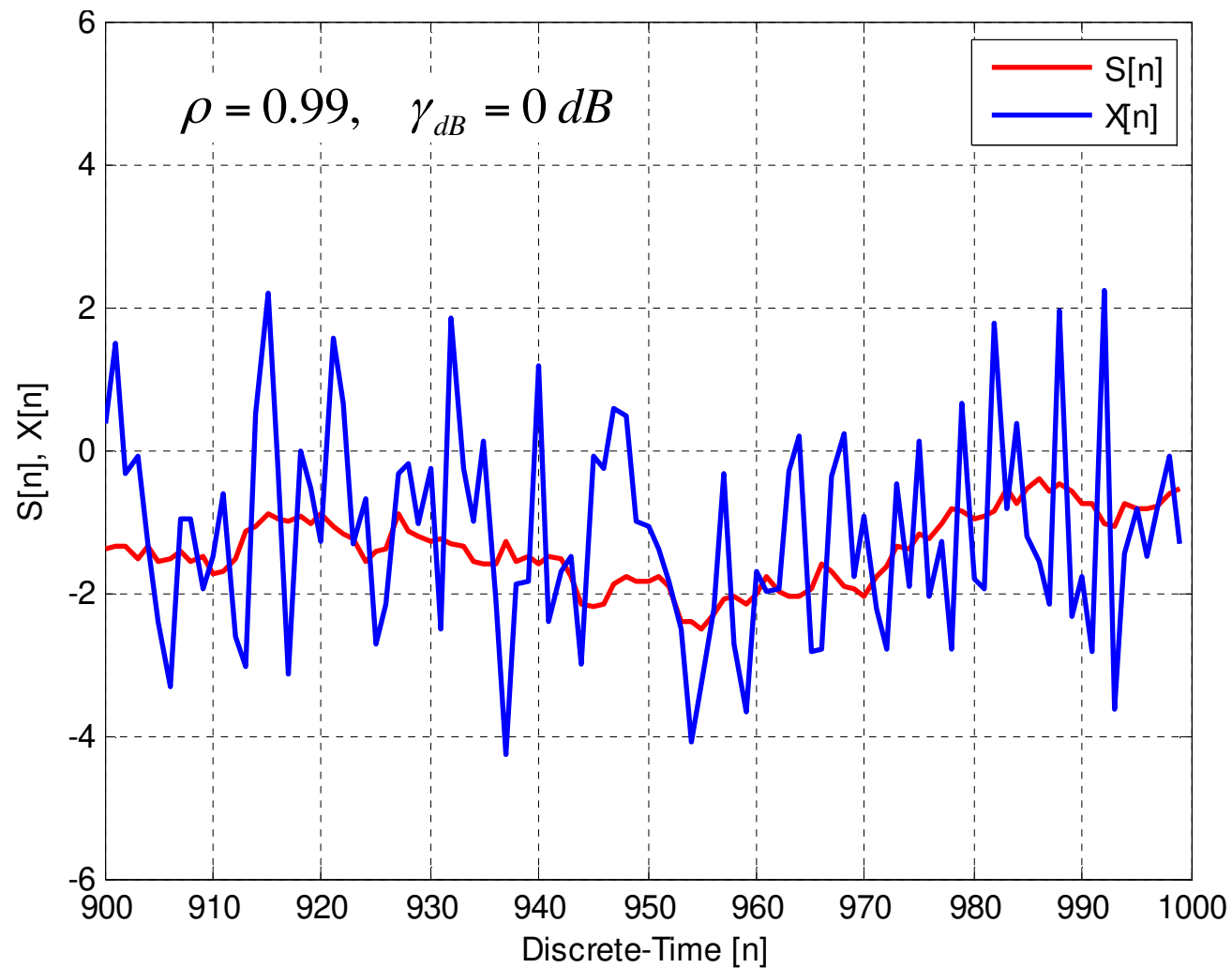
$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



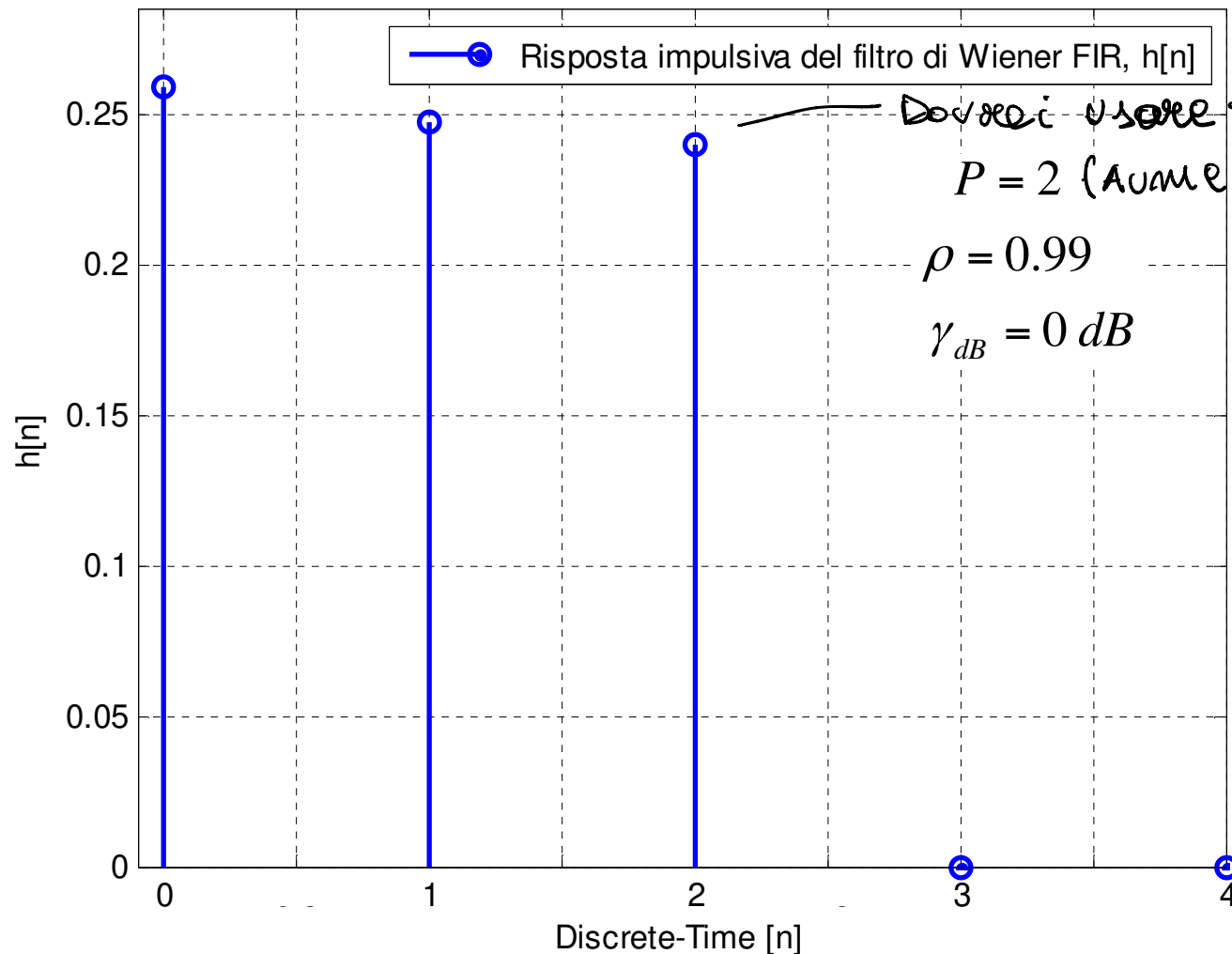
$S[n]$ = Gaussian AR(1) process



$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



Impulse Response (IR) of the FIR(2) Wiener filter

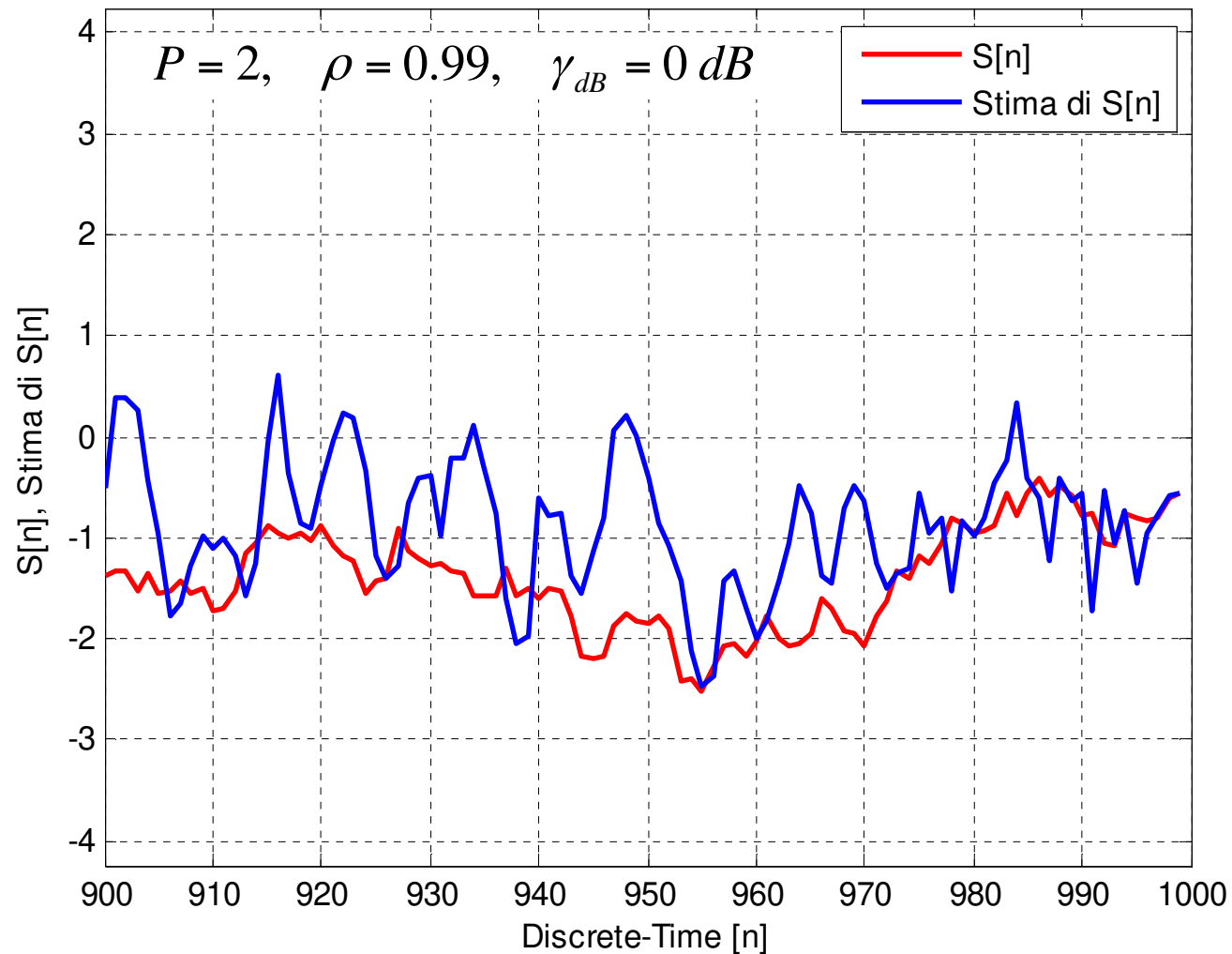


Dovrei usare tanti campioni
 $P = 2$ (Aumentare P)

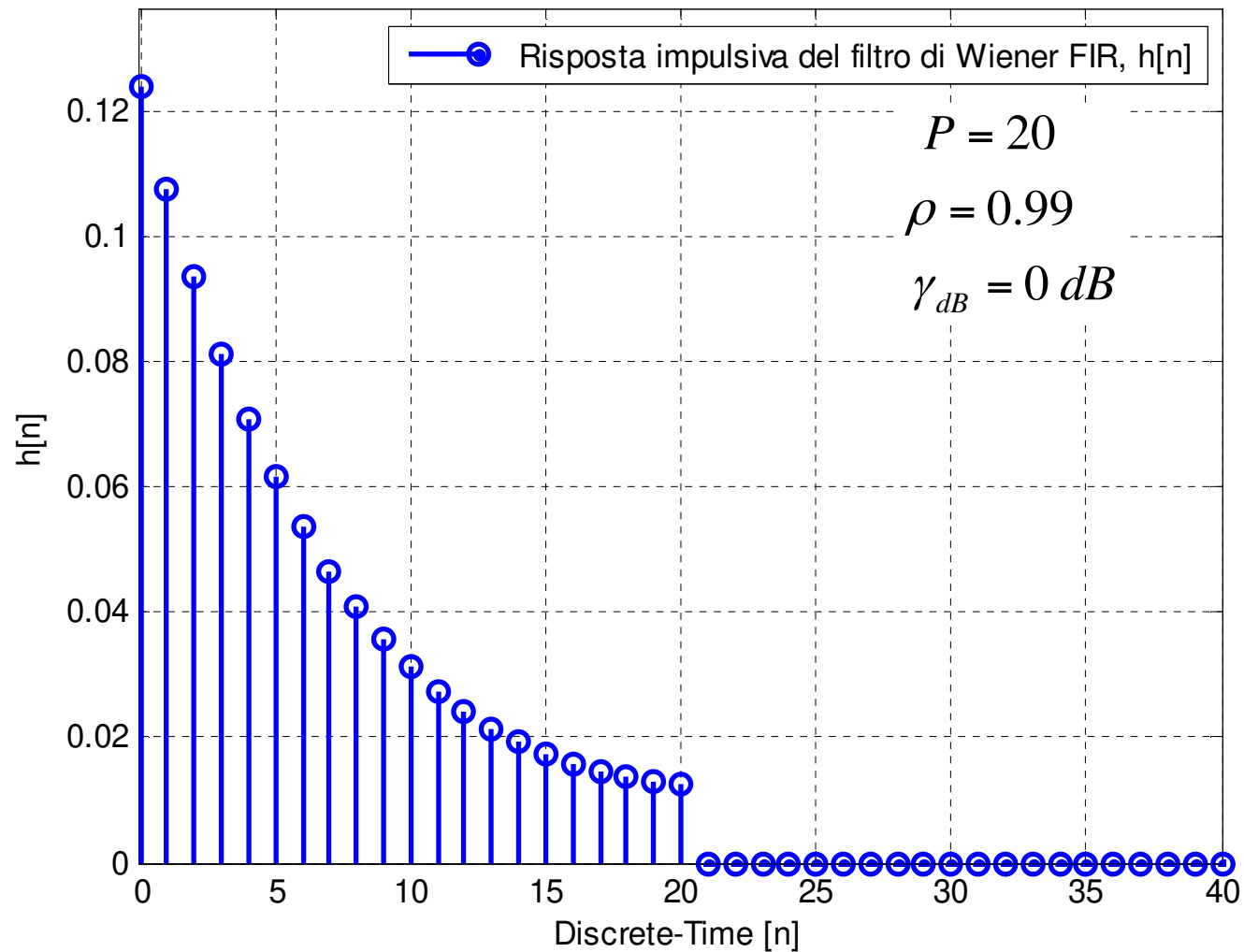
$$\rho = 0.99$$

$$\gamma_{dB} = 0 \text{ dB}$$

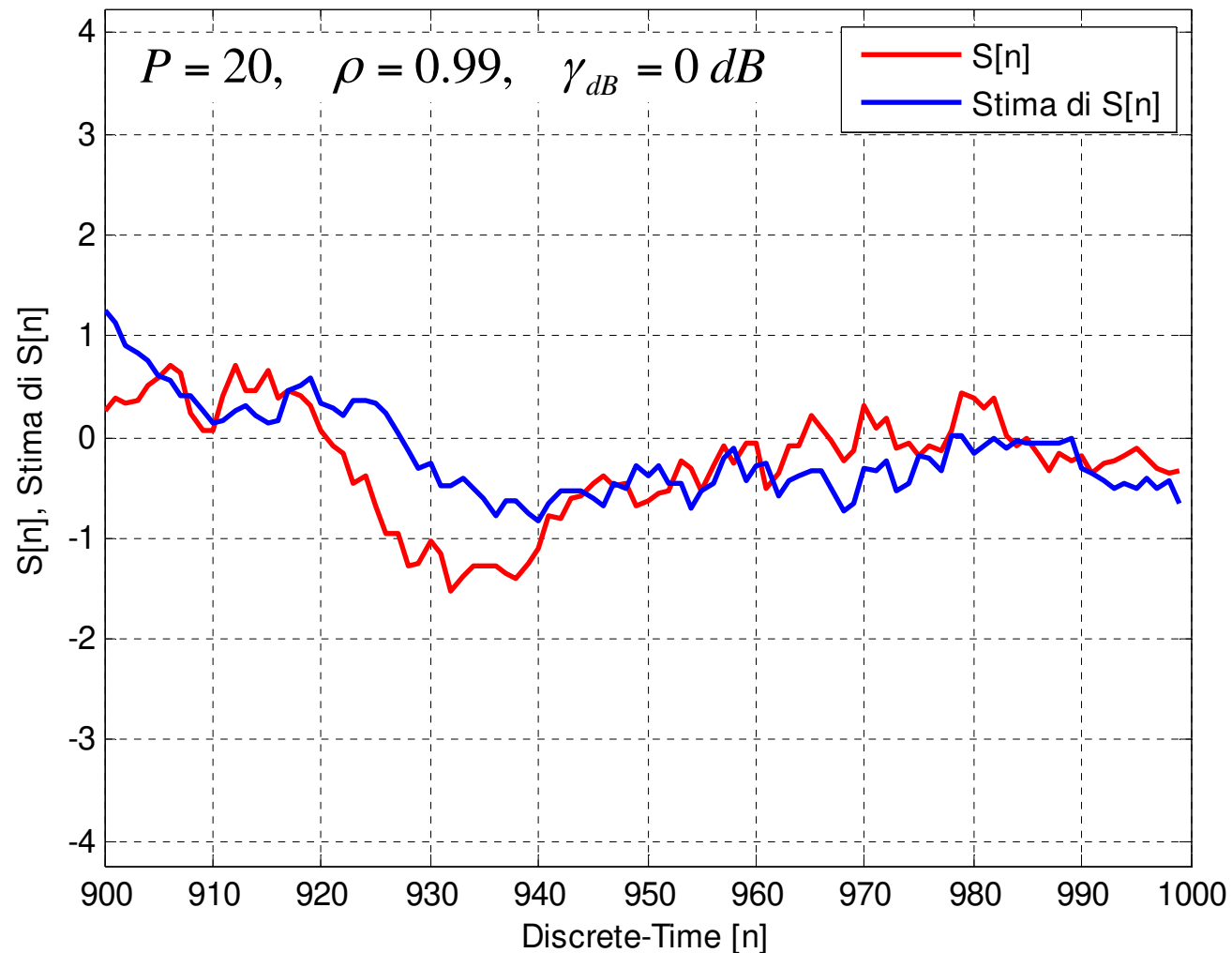
$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



Impulse Response (IR) of the FIR(20) Wiener filter



$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



- The **Mean Squared Error (MSE)** of the estimate achieved by using the FIR Wiener filter of order P is given by:

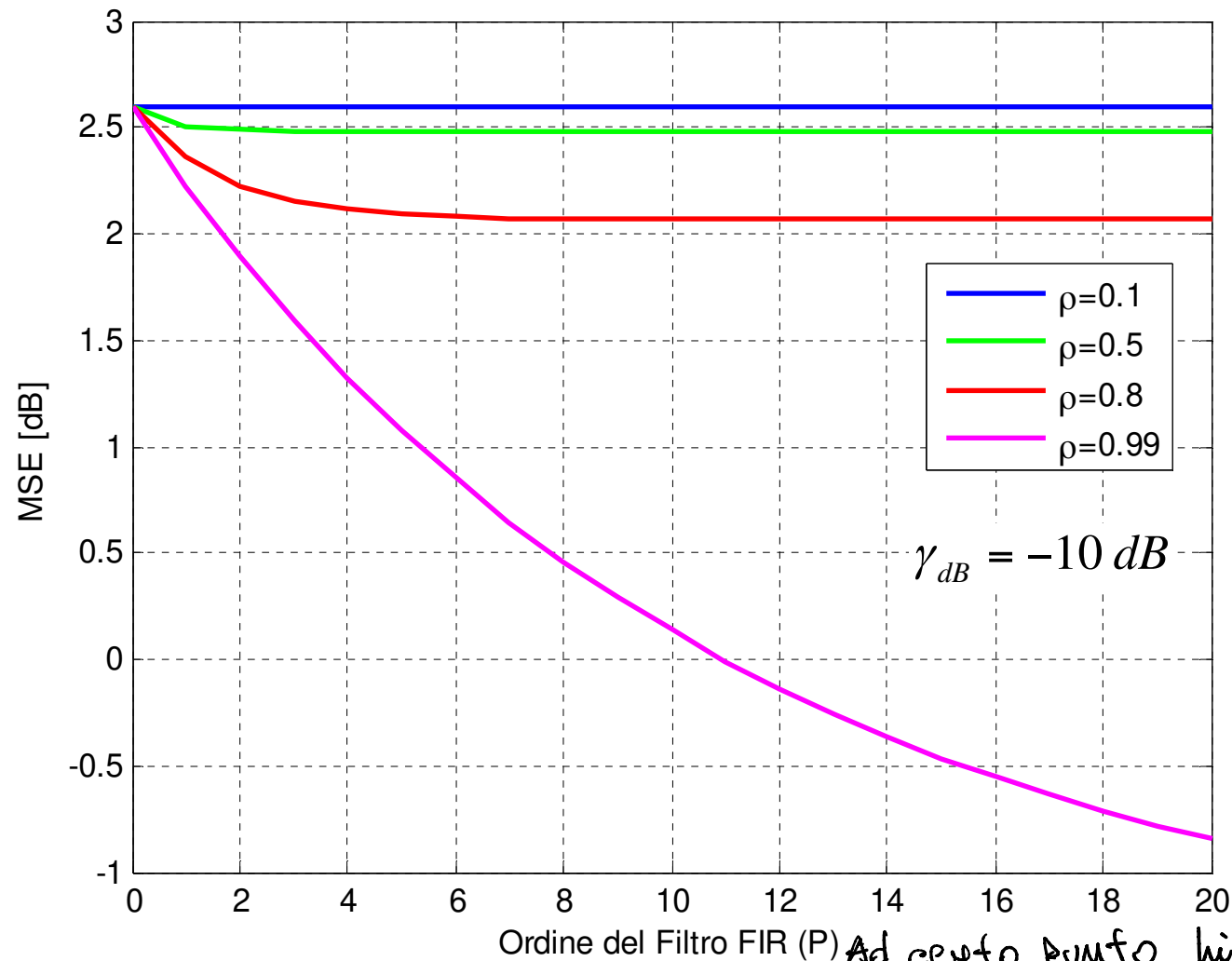
$$\sigma_\varepsilon^2 = R_S[0] - \sum_{k=0}^P h[k] R_S[k] = \sigma_S^2 \left(1 - \sum_{k=0}^P h[k] \rho^k \right)$$

questo dipende da γ quindi SNR

- In the general case of signal and additive noise mutually uncorrelated

- AR(1) signal process

MSE vs P of the FIR(P) Wiener filter when $S[n]$ is an AR(1) process



Ad certo punto tutti i floor
Aumentando P otteniamo meno
Aggiungo campioni incompensati = inutili

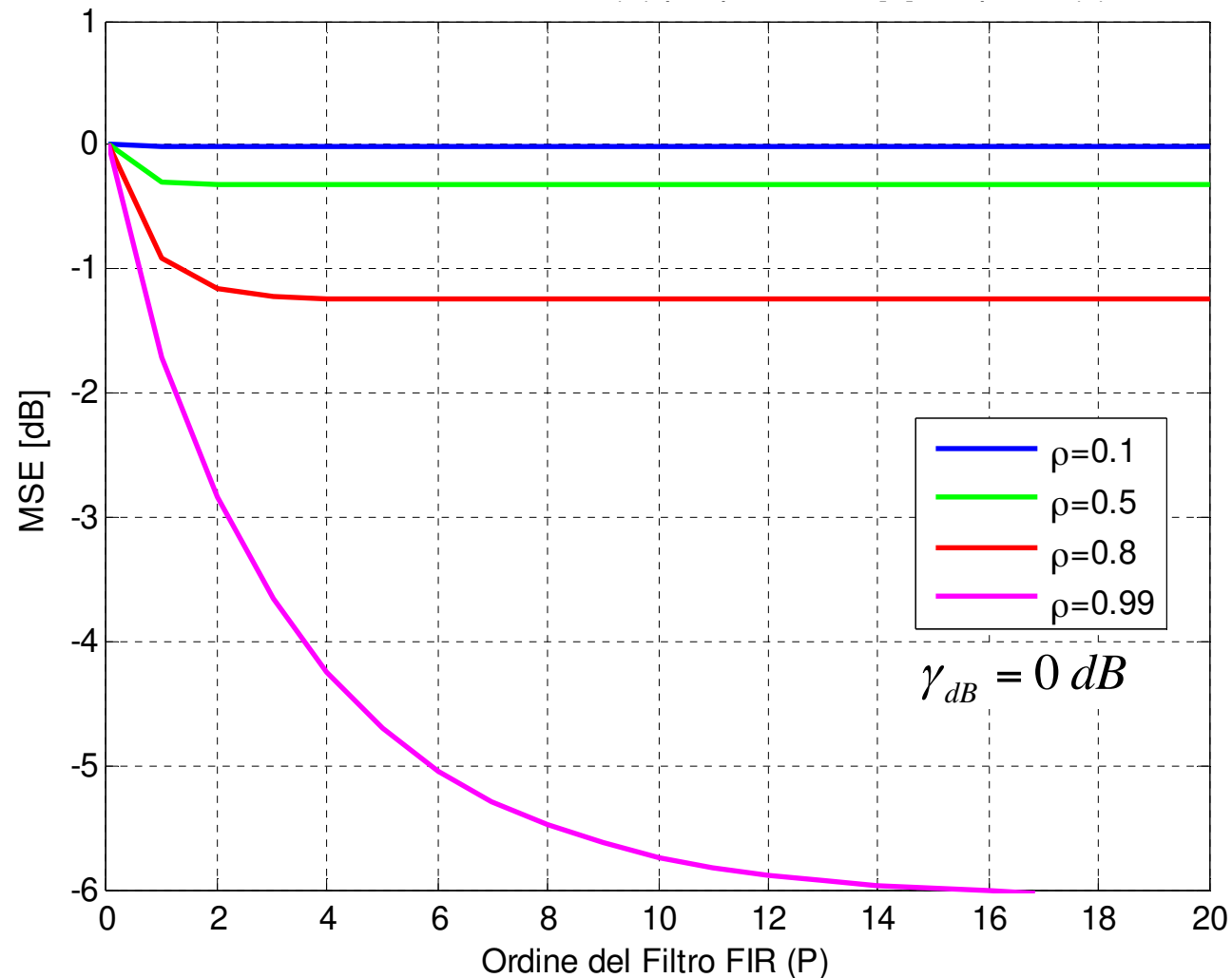


Filtering: FIR Wiener Filter

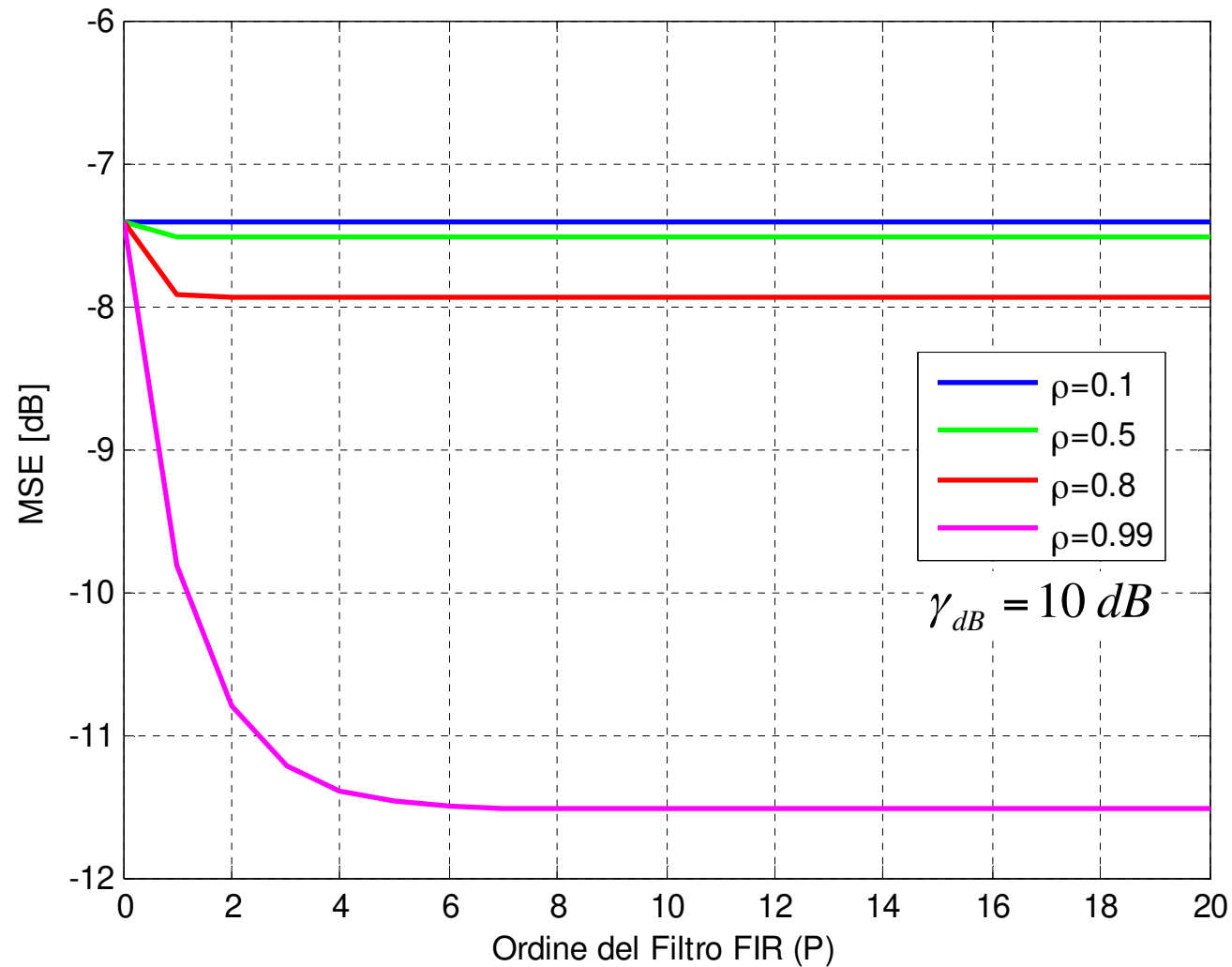
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Caso più favorevole, segnale a ~~se~~-alto

MSE vs P of the FIR(P) Wiener filter when $S[n]$ is an AR(1) process



MSE vs P of the FIR(P) Wiener filter when $S[n]$ is an AR(1) process



- Assume now that we want to estimate the value of the process at discrete-time $n+L$. This is a joint “**Filtering and Prediction**” problem:

$$\hat{S}[n+L] = \sum_{k=0}^P h[k]X[n-k], \quad \text{FIR causal Wiener Filter}$$

- The **prediction error** at the step $n+L$, $\varepsilon[n+L]$, is given by:

$$\varepsilon[n+L] = S[n+L] - \hat{S}[n+L] = S[n+L] - \sum_{k=0}^P h[k]X[n-k]$$

- By applying the **Orthogonality Principle**:

$$E\{\varepsilon[n+L]X[n-m]\} = 0, \quad m = 0, 1, 2, \dots, P$$

- Since $R_{SX}[-(m+L)] = R_S[-(m+L)] = R_S[m+L]$
- The **Wiener-Hopf equation** can be expressed as:

$$\sum_{k=0}^P h[k] R_X[m-k] = R_S[m+L], \quad 0 \leq m \leq P$$

- The difference is here

$$\mathbf{C}_X \mathbf{h} = \mathbf{c}_S$$

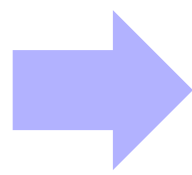
- When $L=0$, we obtain the previous case, i.e. filtering without prediction.

$$\mathbf{C}_X \triangleq \begin{bmatrix} R_X[0] & R_X[-1] & \cdots & R_X[-P] \\ R_X[1] & R_X[0] & \cdots & R_X[-(P-1)] \\ \vdots & \vdots & \ddots & \vdots \\ R_X[P] & R_X[P-1] & \cdots & R_X[0] \end{bmatrix}, \quad \mathbf{c}_S \triangleq \begin{bmatrix} R_S[L] \\ R_S[L+1] \\ \vdots \\ R_S[L+P] \end{bmatrix}, \quad \mathbf{h} \triangleq \begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[P] \end{bmatrix}$$

- **Example:** $S[n]$ AR(1) process and $W[n]$ additive white noise.
- Let us derive the **FIR Wiener filter** of order $P=2$ that also makes a 2-steps prediction ($L=2$).

$$\left\{ \begin{array}{l} X[n] = S[n] + W[n] \\ R_S[m] = \sigma_S^2 \rho^{|m|}, \quad \sigma_S^2 = 2, \quad \rho = 0.8, \quad R_W[m] = \sigma_W^2 \delta[m], \quad \sigma_W^2 = 2 \\ R_X[m] = R_S[m] + R_W[m], \quad R_{SX}[m] = R_S[m] \end{array} \right.$$

$$\Rightarrow \begin{bmatrix} R_X[0] & R_X[-1] & R_X[-2] \\ R_X[1] & R_X[0] & R_X[-1] \\ R_X[2] & R_X[1] & R_X[0] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_S[2] \\ R_S[3] \\ R_S[4] \end{bmatrix}$$



$$2 \cdot \begin{bmatrix} 2 & 0.8 & 0.64 \\ 0.8 & 2 & 0.8 \\ 0.64 & 0.8 & 2 \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \end{bmatrix} = 2 \cdot \begin{bmatrix} 0.64 \\ 0.512 \\ 0.4096 \end{bmatrix}$$

■ The solution is given by:

$$\mathbf{h} = \begin{bmatrix} h[0] \\ h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} 2 & 0.8 & 0.64 \\ 0.8 & 2 & 0.8 \\ 0.64 & 0.8 & 2 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 0.64 \\ 0.512 \\ 0.4096 \end{bmatrix} = \begin{bmatrix} 0.2447 \\ 0.1280 \\ 0.0753 \end{bmatrix}$$

■ The MSE is given by:

$$\begin{aligned} \sigma_{\varepsilon}^2 &= E\{\varepsilon[n+L]S[n+L]\} = R_s[0] - \sum_{k=0}^P h[k]R_s[k+L] = \sigma_s^2 - \sum_{k=0}^2 h[k]\sigma_s^2\rho^{k+2} \\ &= 2 \cdot [1 - 0.2447 \cdot 0.64 - 0.1280 \cdot 0.512 - 0.0753 \cdot 0.4096] = 1.494 \end{aligned}$$

$$\sigma_{\varepsilon}^2 = 1.494 > \sigma_{\varepsilon}^2(L=0) = 0.7647 \text{ Aumento passo} \rightarrow \text{Aumento}$$

- Non causal IIR Wiener filter (the LTI Wiener smoothing filter):

$$\hat{S}[n] = \sum_{k=-\infty}^{+\infty} h[k]X[n-k], \quad \text{Non causal IIR Wiener filter}$$

- The **estimation error** $\varepsilon[n]$ is given by:

$$\varepsilon[n] = S[n] - \hat{S}[n] = S[n] - \sum_{k=-\infty}^{+\infty} h[k]X[n-k]$$

- The coefficients of the IIR Wiener filter $h[n]$ should be obtained by minimizing the MSE, or equivalently, by applying the **Orthogonality Principle**:

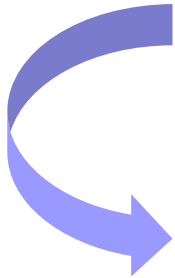
$$\min_{\{h[n]\}} \sigma_{\varepsilon}^2 = E\{\varepsilon^2[n]\} \Rightarrow E\{\varepsilon[n]X[n-m]\} = 0, \quad -\infty < m < +\infty$$

$$E\{\varepsilon[n]X[n-m]\} = R_{SX}[-m] - \sum_{k=-\infty}^{+\infty} h[k]R_X[m-k] = 0, \quad -\infty < m < +\infty$$

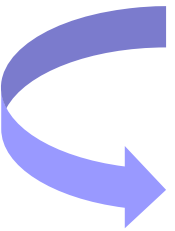


$$\sum_{k=-\infty}^{+\infty} h[k]R_X[m-k] = R_{SX}[-m], \quad -\infty < m < +\infty$$

- The discrete-time Wiener-Hopf equation.
- Since the right and the left sequences are equal for each value of m (and not only for $P+1$ values of m as in the previous case), their Z and Fourier transforms should be equal as well.


$$\sum_{k=-\infty}^{+\infty} h[k]R_X[m-k] = R_{SX}[-m], \quad -\infty < m < +\infty$$

$$h[m] \otimes R_X[m] = R_{SX}[-m] = R_{XS}[m], \quad -\infty < m < +\infty$$


$$H(z)S_X(z) = S_{XS}(z) \Rightarrow H(z) = \frac{S_{XS}(z)}{S_X(z)}$$

$$H(e^{j2\pi f}) = \frac{S_{XS}(e^{j2\pi f})}{S_X(e^{j2\pi f})}, \quad \text{Non causal IIR Wiener Filter}$$

- In the case of signal and additive noise mutually uncorrelated, the frequency response of the **non causal IIR Wiener filter** is:

$$H(e^{j2\pi f}) = \frac{S_{XS}(e^{j2\pi f})}{S_X(e^{j2\pi f})} = \frac{S_S(e^{j2\pi f})}{S_S(e^{j2\pi f}) + S_W(e^{j2\pi f})}$$

- The frequency response can be expressed as function of the “local” Signal-to-Noise power Ratio (SNR) at frequency f :

$$\gamma(e^{j2\pi f}) \triangleq \frac{S_S(e^{j2\pi f})}{S_W(e^{j2\pi f})} \Rightarrow H(e^{j2\pi f}) = \frac{\gamma(e^{j2\pi f})}{\gamma(e^{j2\pi f}) + 1}$$

- $H(\cdot)$ takes values only between 0 and 1, depending on the value of the “local” SNR at that particular frequency:

$$H(e^{j2\pi f}) \in [0,1), \text{ real and even} \Rightarrow h[n] \text{ real and even (i.e. non causal)}$$

$$\gamma(e^{j2\pi f}) = \frac{S_s(e^{j2\pi f})}{S_w(e^{j2\pi f})} \rightarrow +\infty \Rightarrow H(e^{j2\pi f}) = \frac{\gamma(e^{j2\pi f})}{\gamma(e^{j2\pi f}) + 1} \rightarrow 1$$

$$\gamma(e^{j2\pi f}) = \frac{S_s(e^{j2\pi f})}{S_w(e^{j2\pi f})} \rightarrow 0 \Rightarrow H(e^{j2\pi f}) = \frac{\gamma(e^{j2\pi f})}{\gamma(e^{j2\pi f}) + 1} \rightarrow 0$$

- The frequency response of the non causal IIR Wiener filter depends on the shape of the PSD of the signal and the PSD of the noise: the attenuation is larger where the “local” SNR is lower.

■ **Example:** $S[n]$ AR(1) process and $W[n]$ white additive noise.

■ The **non causal IIR Wiener filter** (smoothing) can be derived as follows:

$X[n] = S[n] + W[n]$ prec. e successivi, non è creabile direttamente poiché non causale.

$R_S[m] = \sigma_S^2 \rho^{|m|}, \quad \sigma_S^2 = 2, \quad \rho = 0.8, \quad R_W[m] = \sigma_W^2 \delta[m], \quad \sigma_W^2 = 2$

$R_X[m] = R_S[m] + R_W[m], \quad R_{SX}[m] = R_S[m]$

$S_W(e^{j2\pi f}) = \sigma_W^2$

$S_S(e^{j2\pi f}) = \frac{\sigma_S^2(1 - \rho^2)}{1 + \rho^2 - 2\rho \cos(2\pi f)}$

Usa tutto! (sicuro meglio FIR)

Usa solo questo

$$\gamma(e^{j2\pi f}) = \frac{S_s(e^{j2\pi f})}{S_w(e^{j2\pi f})} = \frac{\sigma_s^2}{\sigma_w^2} \cdot \frac{1 - \rho^2}{1 + \rho^2 - 2\rho \cos(2\pi f)}$$

$$\gamma \triangleq \frac{\sigma_s^2}{\sigma_w^2} \Rightarrow \gamma(e^{j2\pi f}) = \frac{\gamma(1 - \rho^2)}{1 + \rho^2 - 2\rho \cos(2\pi f)}$$

$$H(e^{j2\pi f}) = \frac{\gamma(e^{j2\pi f})}{\gamma(e^{j2\pi f}) + 1} = \frac{\gamma(1 - \rho^2)}{\gamma(1 - \rho^2) + 1 + \rho^2 - 2\rho \cos(2\pi f)}$$

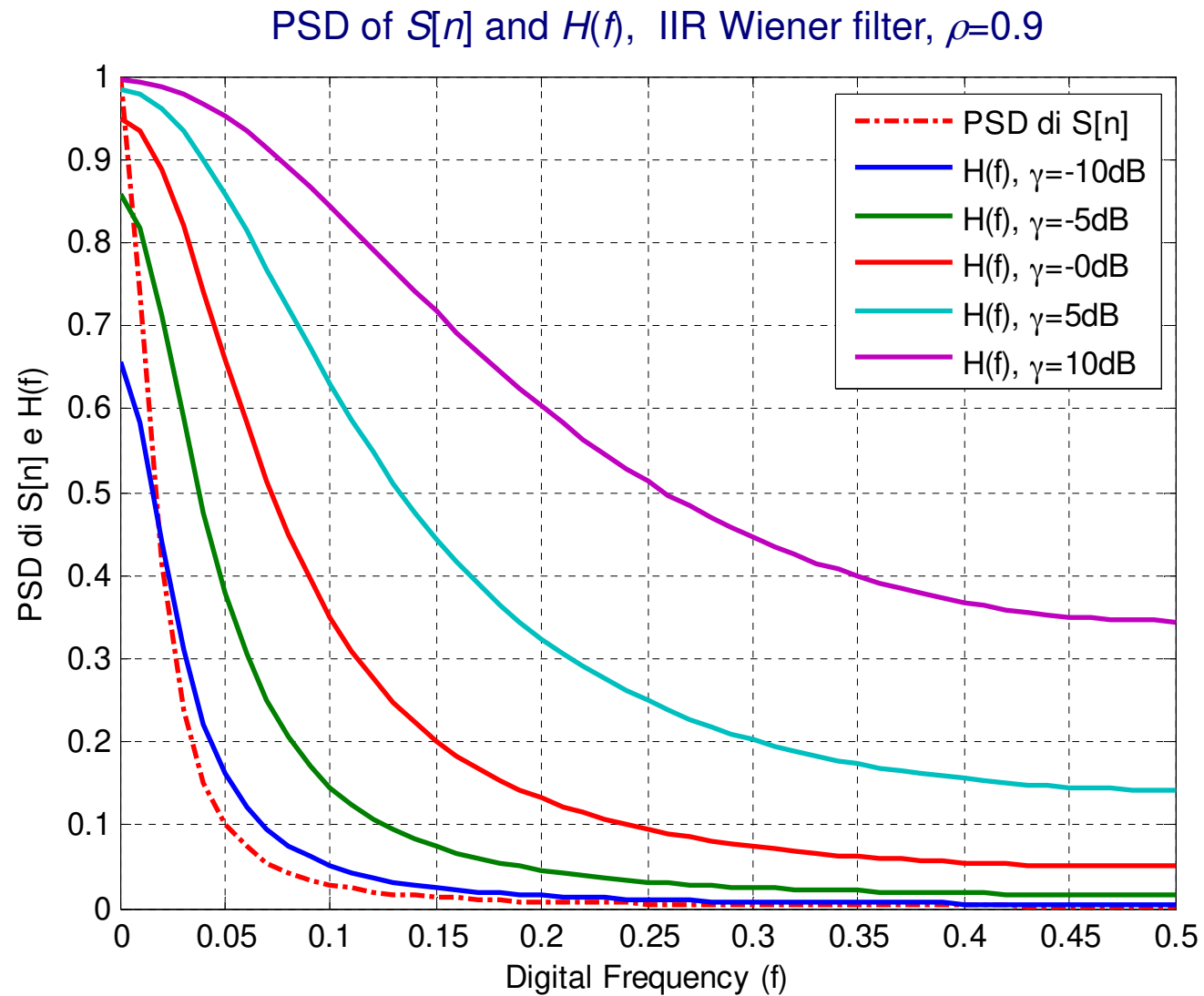
$$H(e^{j2\pi f}) = \frac{\gamma(1-\rho^2)}{\gamma(1-\rho^2) + 1 + \rho^2 - 2\rho \cos(2\pi f)}$$

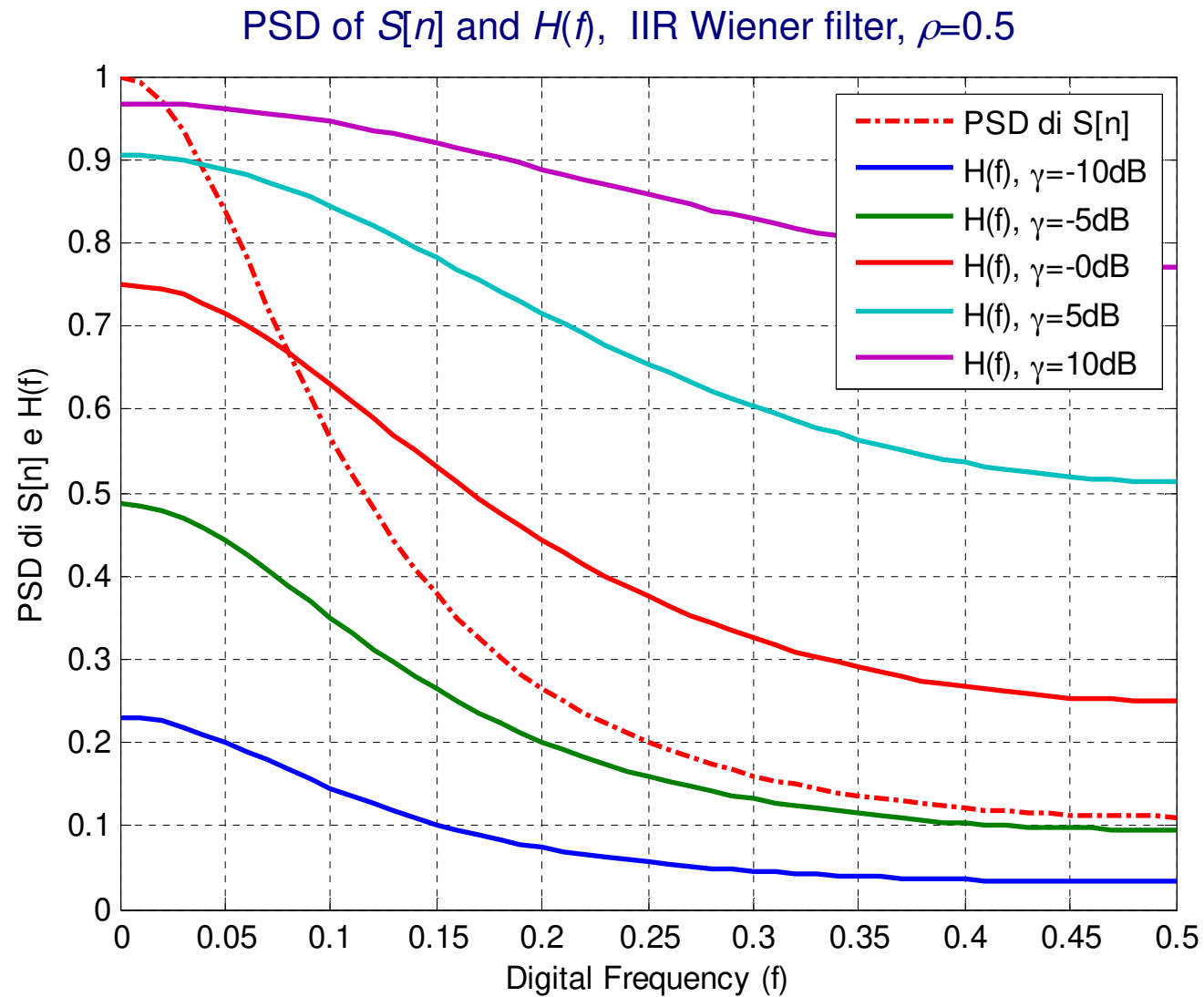
■ In the two limiting cases, we have:

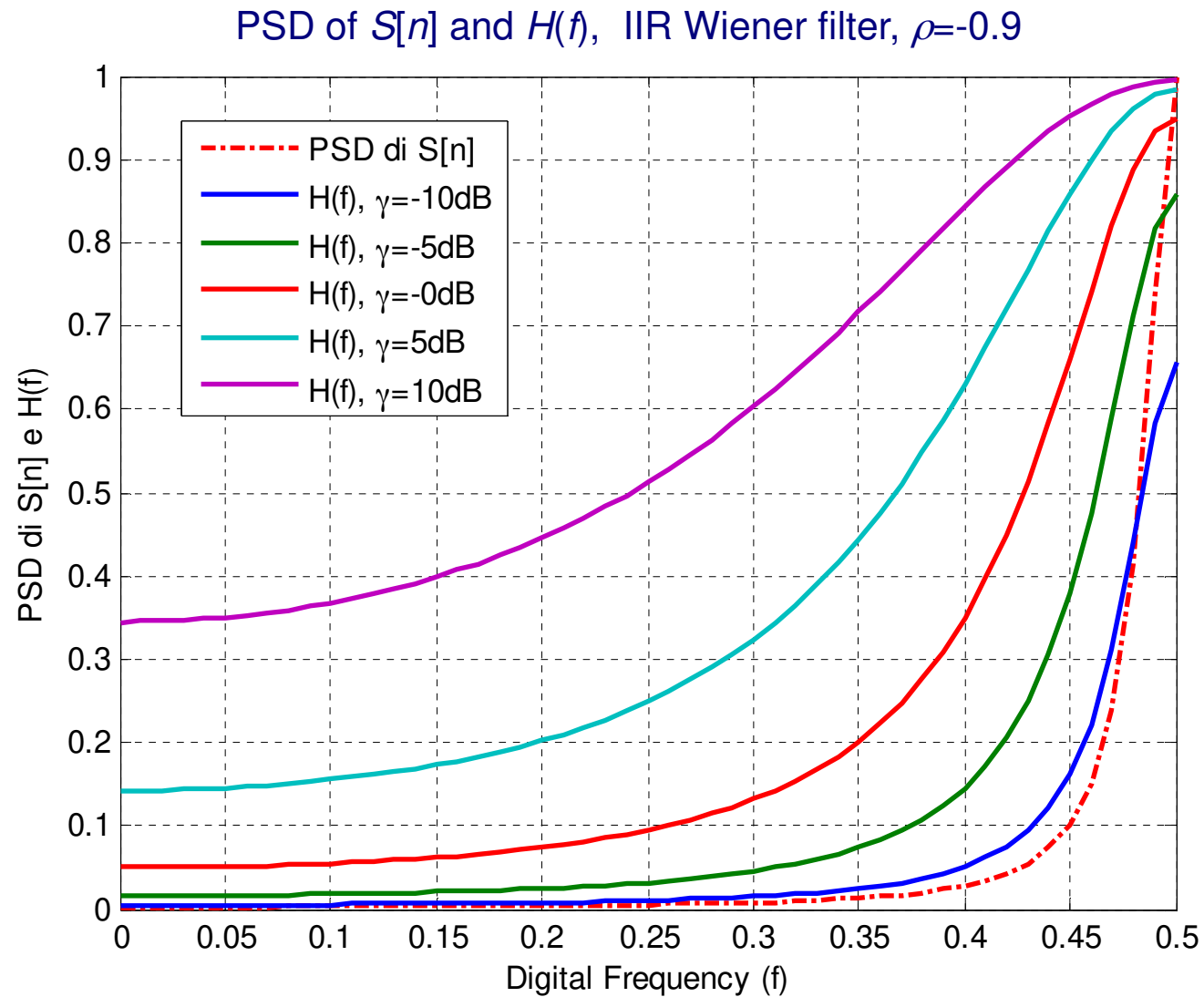
$$\lim_{\gamma \rightarrow \infty} H(e^{j2\pi f}) = \lim_{\gamma \rightarrow \infty} \frac{(1-\rho^2)\gamma}{(1-\rho^2)\gamma + 1 + \rho^2 - 2\rho \cos(2\pi f)} = 1, \quad \forall f$$

$$\lim_{\gamma \rightarrow 0} H(e^{j2\pi f}) = \lim_{\gamma \rightarrow 0} \frac{(1-\rho^2)\gamma}{(1-\rho^2)\gamma + 1 + \rho^2 - 2\rho \cos(2\pi f)} = 0, \quad \forall f$$

$$\text{where } \gamma \triangleq \frac{\sigma_S^2}{\sigma_W^2}$$







- If we want to obtain the system function $H(z)$ and the impulse response $h[n]$ of the **non causal IIR Wiener filter**:

$$H(z) = \frac{S_{XS}(z)}{S_X(z)} = \frac{S_S(z)}{S_S(z) + S_W(z)} = \frac{\sigma_I^2 L(z) L(1/z)}{\sigma_I^2 L(z) L(1/z) + \sigma_W^2}$$

$$\sigma_s^2 = \frac{\sigma_I^2}{1 - \rho^2}, \quad \gamma = \frac{\sigma_s^2}{\sigma_w^2}, \quad L(z) = \frac{1}{1 - \rho z^{-1}}$$

$$H(z) = \frac{\frac{\sigma_s^2(1 - \rho^2)}{(1 - \rho z^{-1})(1 - \rho z)}}{\frac{\sigma_s^2(1 - \rho^2)}{(1 - \rho z^{-1})(1 - \rho z)} + \sigma_w^2} = \frac{\gamma(1 - \rho^2)}{\gamma(1 - \rho^2) + (1 - \rho z^{-1})(1 - \rho z)}$$

$$= \frac{\beta}{(1 - \alpha z^{-1})(1 - \alpha z)} \quad \text{with } \alpha \text{ and } \beta \text{ to be determined}$$

- α is the root within the unit circle of the polynomial at the denominator.
The two roots α and $1/\alpha$ of the denominator can be easily obtained as:

$$\begin{aligned}\gamma(1-\rho^2) + (1-\rho z^{-1})(1-\rho z) &= \gamma(1-\rho^2) + 1 + \rho^2 - \rho z^{-1} - \rho z \\ &= -\rho z^{-1} \left(z^2 - \frac{1+\rho^2+\gamma(1-\rho^2)}{\rho} z + 1 \right) = 0\end{aligned}$$

$$\Rightarrow p_{1,2} = \frac{1+\rho^2+\gamma(1-\rho^2) \pm \sqrt{[1+\rho^2+\gamma(1-\rho^2)]^2 - 4\rho^2}}{2\rho}$$

$$\Rightarrow \alpha = p_2 = \frac{1+\rho^2+\gamma(1-\rho^2) - \sqrt{[1+\rho^2+\gamma(1-\rho^2)]^2 - 4\rho^2}}{2\rho}$$

- Once α has been determined, β can be obtained by posing the equality between the two polynomials, for example for $z=1$:

$$\left. \frac{\beta}{(1 - \alpha z^{-1})(1 - \alpha z)} \right|_{z=1} = \left. \frac{\gamma(1 - \rho^2)}{\gamma(1 - \rho^2) + (1 - \rho z^{-1})(1 - \rho z)} \right|_{z=1}$$

$$\Rightarrow \beta = \frac{\gamma(1 - \rho^2)(1 - \alpha)^2}{\gamma(1 - \rho^2) + (1 - \rho)^2}$$

- Having determined $H(z)$, calculating the inverse Z-transform we can obtain the impulse response of the **non casual IIR Wiener filter**:

$$H(z) = \frac{\beta}{(1 - \alpha z^{-1})(1 - \alpha z)} \xrightarrow{ZT^{-1}} h[n] = ?$$

- Since we know that for an AR(1) process:

$$R_s[m] = \sigma_s^2 \rho^{|m|} \quad \stackrel{ZT}{\Leftrightarrow} \quad S_s(z) = \frac{\sigma_s^2 (1 - \rho^2)}{(1 - \rho z^{-1})(1 - \rho z)}$$

- Hence if we set:

$$\beta \triangleq \sigma_s^2 (1 - \rho^2), \quad \alpha \triangleq \rho \quad \rightarrow \quad \rho = \alpha, \quad \sigma_s^2 = \frac{\beta}{1 - \alpha^2}$$

- it follows immediately that:

$$h[n] = \frac{\beta}{1 - \alpha^2} \alpha^{|n|} \quad \stackrel{ZT}{\Leftrightarrow} \quad H(z) = \frac{\beta}{(1 - \alpha z^{-1})(1 - \alpha z)}, \quad \alpha < |z| < \frac{1}{\alpha}$$

- The signal estimate can be expressed as follows:

$$\hat{S}[n] = \sum_{k=-\infty}^{+\infty} h[k]X[n-k] = \frac{\beta}{1-\alpha^2} \sum_{k=-\infty}^{+\infty} \alpha^{|k|} X[n-k]$$

- In practice, only an approximation of this estimate can be implemented, and not in real-time.
- We have first to apply a truncation of the observation window, in order to have a window of finite length $2P+1$, and then to introduce a delay of P , to make the filter causal (and then feasible).

$$P: |h[\pm P]| \ll |h[0]| \Rightarrow \alpha^P \ll 1 \text{ (Remoto realizable)}$$

■ Assume for example that for our scope: $\alpha^P \ll 1 \Leftrightarrow \alpha^P = 10^{-3}$

■ Then we get:

$$\text{Log}_{10}(\alpha^P) = \text{Log}_{10}(10^{-3}) \Rightarrow P = -\frac{3}{\text{Log}_{10}\alpha} = \frac{3}{\text{Log}_{10}(1/\alpha)}$$

■ Once we have determined P , we have to introduce a delay P to make the filter causal:

$$\hat{S}[n] \cong \frac{\beta}{1-\alpha^2} \sum_{k=-P}^P \alpha^{|k|} X[n-k] \Rightarrow \hat{S}[n-P] \cong \frac{\beta}{1-\alpha^2} \sum_{k=-P}^P \alpha^{|k|} X[n-P-k]$$

$$\hat{S}[n-P] \cong \frac{\beta}{1-\alpha^2} \sum_{k=-P}^P \alpha^{|k|} X[n-(k+P)] = \frac{\beta}{1-\alpha^2} \sum_{i=0}^{2P} \alpha^{|i-P|} X[n-i]$$

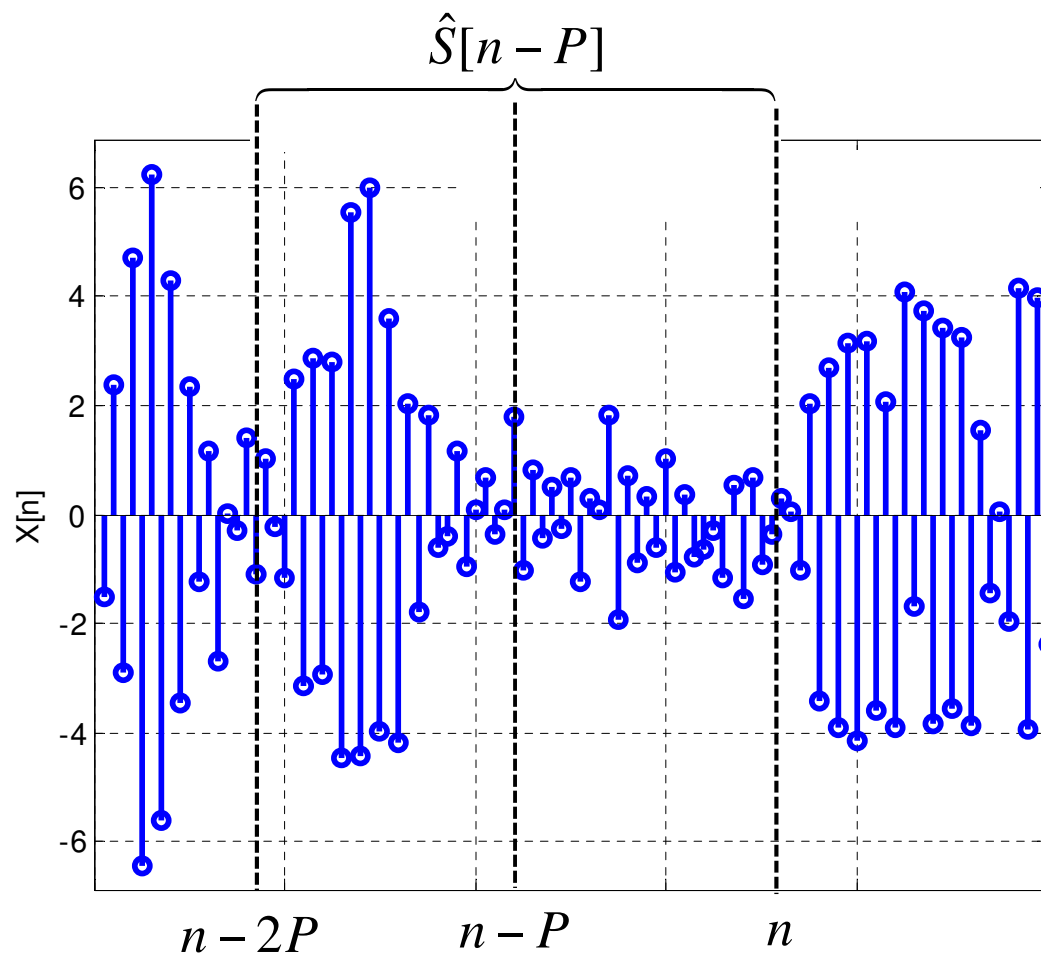
Тогда для causality -> имеем so много processing 51

Smoothing: non causal IIR Wiener filter

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se il ritardo me ne sbatto ci sta u scello

$$\hat{S}[n-P] = \frac{\beta}{1-\alpha^2} \sum_{i=0}^{2P} \alpha^{|i-P|} X[n-i]$$



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■ In the particular case: $\sigma_s^2 = 2$, $\rho = 0.8$, $\sigma_w^2 = 2 \Rightarrow \gamma = 1$

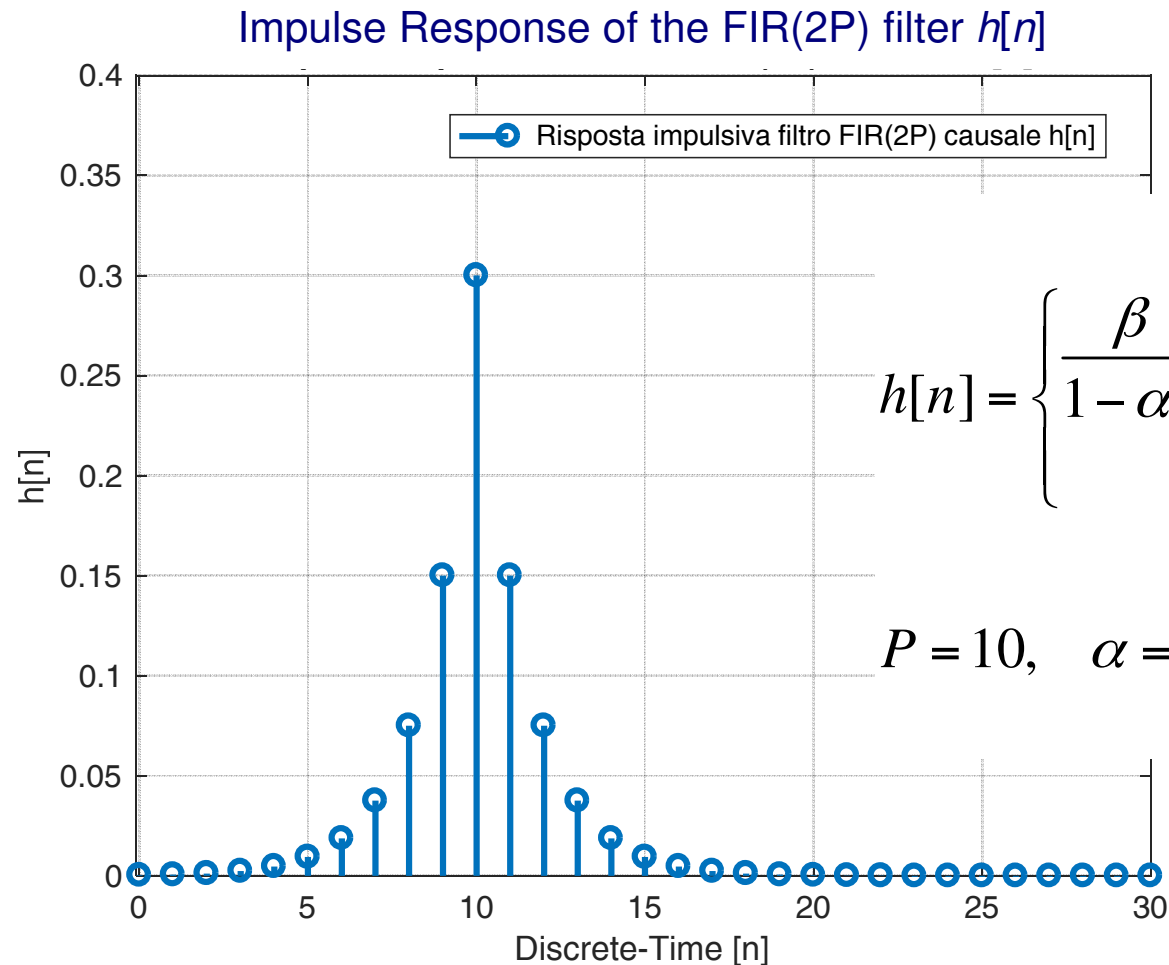
$$p_{1,2} = \frac{1 + 0.8^2 + 1 \cdot (1 - 0.8^2) \pm \sqrt{[1 + 0.8^2 + 1 \cdot (1 - 0.8^2)]^2 - 4 \cdot 0.8^2}}{2 \cdot 0.8} = (0.5, 2)$$

$$\Rightarrow \alpha = p_2 = 0.5, \quad 1/\alpha = p_1 = 2$$

$$\Rightarrow \beta = \frac{1 \cdot (1 - 0.8^2)(1 - 0.5)^2}{1 \cdot (1 - 0.8^2) + (1 - 0.8)^2} = 0.225$$

$$\Rightarrow H(z) = \frac{0.225}{(1 - 0.5z^{-1})(1 - 0.5z)} \Rightarrow h[n] = \frac{0.225}{1 - 0.5^2} 0.5^{|n|} = 0.3 \cdot 0.5^{|n|}$$

- The causal FIR(2P) filter that approximates the behavior of the **non causal IIR Wiener filter** has impulse response:



$$h[n] = \begin{cases} \frac{\beta}{1 - \alpha^2} \cdot \alpha^{|n-P|}, & 0 \leq n \leq 2P \\ 0, & \text{otherwise} \end{cases}$$

$$P = 10, \quad \alpha = 0.5, \quad \beta = 0.225$$

- The MSE of the **non causal IIR Wiener filter** is given by:

$$\begin{aligned}\sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{\varepsilon[n]S[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{S^2[n]\} - \sum_{k=-\infty}^{+\infty} h[k]E\{S[n]X[n-k]\} \\ &= R_S[0] - \sum_{k=-\infty}^{+\infty} h[k]R_{SX}[-k] \\ &= R_S[0] - \sum_{k=-\infty}^{+\infty} h[k]R_S[k]\end{aligned}$$

- **Example:** $S[n]$ AR(1) process and $W[n]$ additive white noise uncorrelated with the signal.

$$R_S[m] = \sigma_s^2 \rho^{|m|}, \quad h[n] = \frac{\beta}{1 - \alpha^2} \alpha^{|n|}$$

$$\begin{aligned} \sigma_\varepsilon^2 &= R_S[0] - \sum_{k=-\infty}^{+\infty} h[k] R_S[k] = \sigma_s^2 - \frac{\beta}{1 - \alpha^2} \sum_{k=-\infty}^{+\infty} \alpha^{|k|} \sigma_s^2 \rho^{|k|} \\ &= \sigma_s^2 \left(1 - \frac{\beta}{1 - \alpha^2} \sum_{k=-\infty}^{+\infty} (\alpha \rho)^{|k|} \right) = \sigma_s^2 \left(1 - \frac{\beta}{1 - \alpha^2} \cdot \frac{1 + \alpha \rho}{1 - \alpha \rho} \right) \end{aligned}$$

$$\text{since: } \sum_{k=-\infty}^{+\infty} (\alpha \rho)^{|k|} = 2 \cdot \sum_{k=0}^{+\infty} (\alpha \rho)^k - 1 = 2 \cdot \frac{1}{1 - \alpha \rho} - 1 = \frac{1 + \alpha \rho}{1 - \alpha \rho}$$

- In the particular case: $\sigma_s^2 = 2$, $\rho = 0.8$, $\sigma_w^2 = 2 \Rightarrow \gamma = 1$

$$\sigma_\varepsilon^2 = \sigma_s^2 \left(1 - \frac{\beta}{1 - \alpha^2} \cdot \frac{1 + \alpha\rho}{1 - \alpha\rho} \right) = 2 \left(1 - \frac{0.225}{1 - 0.5^2} \cdot \frac{1 + 0.5 \cdot 0.8}{1 - 0.5 \cdot 0.8} \right) = 0.6$$

- As expected, the MSE of the non causal IIR filter is lower than the MSE of the causal FIR(2) filter previously derived (this is always true, not only in this particular case):

$$\sigma_{\varepsilon, FIR(2)}^2 = 0.7647 > \sigma_{\varepsilon, IIRnc}^2 = 0.6$$

■ Causal IIR Wiener filter:

$$\hat{S}[n] = \sum_{k=0}^{+\infty} h[k]X[n-k], \quad \text{causal IIR Wiener filter}$$

■ The **estimation error** $\varepsilon[n]$ is given by:

$$\varepsilon[n] = S[n] - \hat{S}[n] = S[n] - \sum_{k=0}^{+\infty} h[k]X[n-k]$$

■ The coefficients of the causal IIR Wiener filter $h[n]$ should be obtained by minimizing the MSE, or equivalently, by applying the **Orthogonality Principle**:

$$\min_{\{h[n]\}} \sigma_{\varepsilon}^2 = E\{\varepsilon^2[n]\} \Rightarrow E\{\varepsilon[n]X[n-m]\} = 0, \quad 0 \leq m < +\infty$$

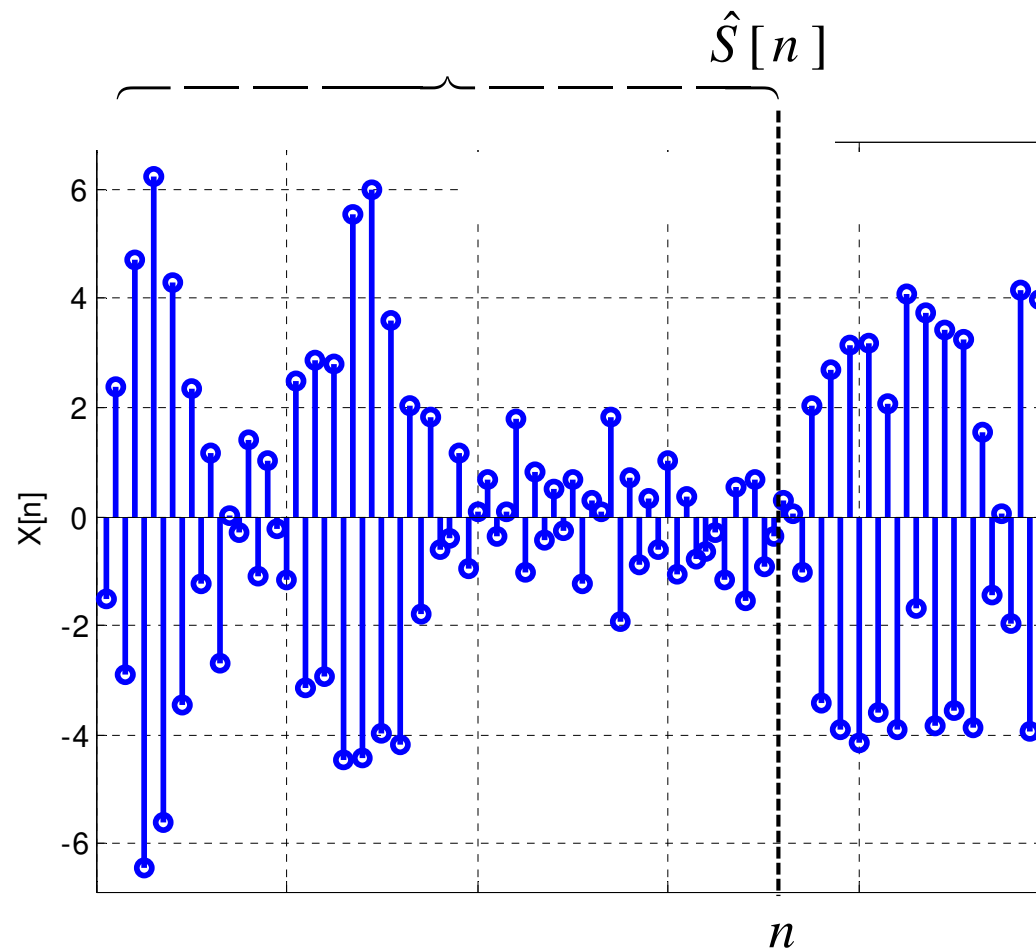
$$E\{\varepsilon[n]X[n-m]\} = R_{SX}[m] - \sum_{k=0}^{+\infty} h[k]R_X[m-k] = 0, \quad 0 \leq m < +\infty$$



$$\sum_{k=0}^{+\infty} h[k]R_X[m-k] = R_{SX}[-m], \quad 0 \leq m < +\infty$$

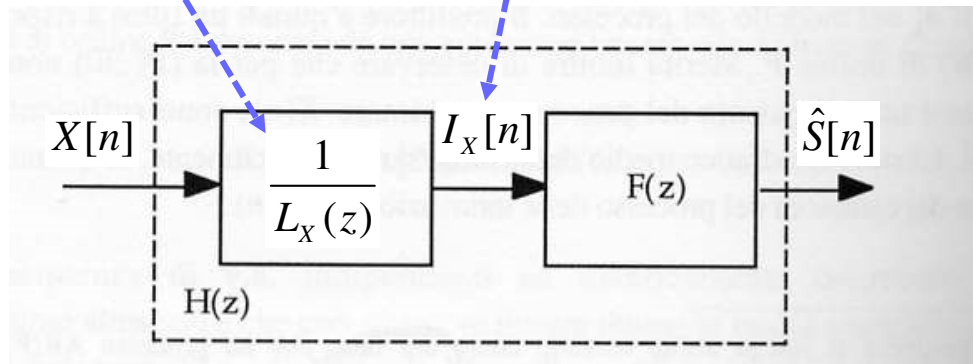
- The discrete-time Wiener-Hopf equation.
- Since the left-hand and right-hand sequences are NOT equal for each value of m , but only for $m \geq 0$, their Z transform will be different. We cannot solve this problem simply going to the Z domain, as in the case of the non causal IIR filter.

$$\hat{S}[n] = \sum_{k=0}^{+\infty} h[k]X[n-k], \quad \text{causal IIR Wiener filter}$$



- To find $H(z)$ can follow a procedure similar to the one we adopted to derive the 1-step linear predictor for a regular process, i.e. we derive first the optimal causal IIR filter that operates on the whitened data $I_X[n]$:

$$\gamma_X[n] \stackrel{ZT}{\Leftrightarrow} \Gamma_X(z) = \frac{1}{L_X(z)} \quad I_X[n] = \gamma_X[n] \otimes X[n] = \sum_{k=0}^{+\infty} \gamma_X[k] X[n-k]$$

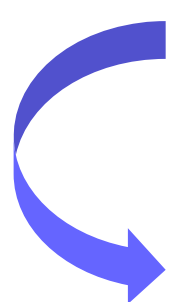


$$\hat{S}[n] = \sum_{k=0}^{+\infty} h[k] X[n-k] = \sum_{k=0}^{+\infty} f[k] I_X[n-k]$$

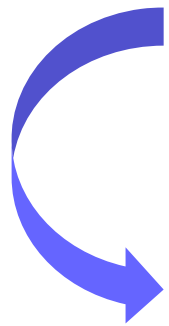
- The optimal linear filter that process the whitened data can be obtained, as usual, by imposing the orthogonality between the error and the samples of the innovation process used for the estimate:

$$E\{\varepsilon[n]I_X[n-m]\} = 0, \quad m \geq 0$$

$$E\left\{\left(S[n] - \hat{S}[n]\right)I_X[n-m]\right\} = R_{SI}[-m] - E\left\{\sum_{k=0}^{+\infty} f[k]I_X[n-k]I_X[n-m]\right\}$$


$$= R_{SI}[-m] - \sum_{k=0}^{+\infty} f[k]R_I[m-k] = 0, \quad 0 \leq m < +\infty$$

$$\sum_{k=0}^{+\infty} f[k]\sigma_I^2\delta[m-k] = R_{SI}[-m], \quad 0 \leq m < +\infty$$



$$\sigma_I^2 f[m] = R_{SI}[-m], \quad 0 \leq m < +\infty$$

$$f[n] = \begin{cases} \frac{1}{\sigma_I^2} R_{SI}[-n], & n \geq 0 \\ 0, & n < 0 \end{cases}$$

■ $R_{SI}[-m]$ can be derived, using this relation: $I_X[n] = \gamma_X[n] \otimes X[n]$

$$\begin{aligned} R_{SI}[-m] &= E\{S[n]I_X[n-m]\} = E\left\{S[n]\sum_{k=0}^{+\infty}\gamma_X[k]X[n-m-k]\right\} \\ &= \sum_{k=0}^{+\infty}\gamma_X[k]E\{S[n]X[n-m-k]\} = \sum_{k=0}^{+\infty}\gamma_X[k]R_{SX}[-(m+k)] \\ &= \sum_{k=0}^{+\infty}\gamma_X[k]R_{XS}[m+k] = R_{XS}[m] \otimes \gamma_X[-m] \end{aligned}$$

$$R_{SI}[-m] = R_{XS}[m] \otimes \gamma_X[-m] \quad \stackrel{ZT}{\Leftrightarrow} \quad ZT\{R_{SI}[-m]\} = S_{XS}(z)\Gamma_X(1/z) = \frac{S_{XS}(z)}{L_X(1/z)}$$

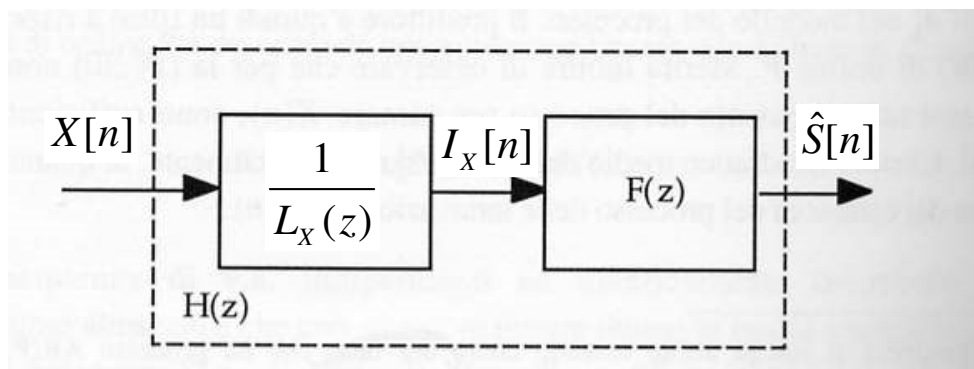
where we used the fact that $\gamma_X[-m] \stackrel{ZT}{\Leftrightarrow} \Gamma_X(1/z)$

$$ZT\{\gamma_X[-m]\} = \sum_{m=-\infty}^{+\infty} \gamma_X[-m]z^{-m} = \sum_{k=-\infty}^{+\infty} \gamma_X[k]z^k = \sum_{k=-\infty}^{+\infty} \gamma_X[k](1/z)^{-k} = \Gamma_X(1/z)$$

- The system function of the optimal linear IIR Wiener filter that operates on the whitened data can be obtained by taking only the causal part, that is the part for $n \geq 0$ that correspond to the singularities within the unit circle:

$$f[n] = \frac{1}{\sigma_I^2} R_{SI}[-n]u[n] \quad \stackrel{ZT}{\Leftrightarrow} \quad F(z) = \frac{1}{\sigma_I^2} \left[\frac{S_{XS}(z)}{L_X(1/z)} \right]_+$$

- The system function of the optimal linear filter that processes the data $X[n]$ can be obtained as the cascade of the whitening filter and of $F(z)$:



$$\hat{S}[n] = \sum_{k=0}^{+\infty} h[k]X[n-k]$$

$$H(z) = \frac{F(z)}{L_X(z)} = \frac{1}{\sigma_I^2 L_X(z)} \left[\frac{S_{XS}(z)}{L_X(1/z)} \right]_+$$

- If signal and noise processes are uncorrelated:

$$H(z) = \frac{1}{\sigma_I^2 L_X(z)} \left[\frac{S_{XS}(z)}{L_X(1/z)} \right]_+ = \frac{1}{\sigma_I^2 L_X(z)} \left[\frac{S_S(z)}{L_X(1/z)} \right]_+$$

- The MSE of the **causal IIR Wiener filter** is given by:

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{\varepsilon[n]S[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} \\ &= E\{S^2[n]\} - \sum_{k=0}^{+\infty} h[k] E\{S[n]X[n-k]\} \\ &= R_S[0] - \sum_{k=0}^{+\infty} h[k] R_{SX}[-k] \\ &= R_S[0] - \sum_{k=0}^{+\infty} h[k] R_S[k] \end{aligned}$$

- **Example:** $S[n]$ AR(1) process and $W[n]$ white additive noise.
- The **causal IIR Wiener filter** can be obtained as follows:

$$\left\{ \begin{array}{l} X[n] = S[n] + W[n] \\ R_S[m] = \sigma_S^2 \rho^{|m|}, \quad \sigma_S^2 = 2, \quad \rho = 0.8, \quad R_W[m] = \sigma_W^2 \delta[m], \quad \sigma_W^2 = 2 \\ R_X[m] = R_S[m] + R_W[m], \quad R_{SX}[m] = R_S[m] \end{array} \right.$$

$$R_W[m] = \sigma_W^2 \delta[m] \quad \stackrel{ZT}{\Leftrightarrow} \quad S_W(z) = \sigma_W^2$$

$$R_S[m] = \sigma_S^2 \rho^{|m|} \quad \stackrel{ZT}{\Leftrightarrow} \quad S_S(z) = \frac{\sigma_S^2 (1 - \rho^2)}{(1 - \rho z^{-1})(1 - \rho z)}$$

$$S_{XS}(z) = S_S(z) = \frac{\sigma_s^2(1-\rho^2)}{(1-\rho z^{-1})(1-\rho z)}$$

$$\begin{aligned} S_X(z) &= S_S(z) + S_W(z) = \frac{\sigma_s^2(1-\rho^2)}{(1-\rho z^{-1})(1-\rho z)} + \sigma_w^2 \\ &= \sigma_w^2 \cdot \frac{\gamma(1-\rho^2) + (1-\rho z^{-1})(1-\rho z)}{(1-\rho z^{-1})(1-\rho z)} \\ &= \sigma_w^2 \cdot \frac{\gamma(1-\rho^2)}{\beta} \cdot \frac{(1-\alpha z^{-1})(1-\alpha z)}{(1-\rho z^{-1})(1-\rho z)} \end{aligned}$$

■ where we used the following relation that we had previously derived:

$$H_{IIRnc}(z) = \frac{\gamma(1-\rho^2)}{\gamma(1-\rho^2) + (1-\rho z^{-1})(1-\rho z)} = \frac{\beta}{(1-\alpha z^{-1})(1-\alpha z)}$$



$$\left\{ \begin{array}{l} \alpha = \frac{1 + \rho^2 + \gamma(1 - \rho^2) - \sqrt{[1 + \rho^2 + \gamma(1 - \rho^2)]^2 - 4\rho^2}}{2\rho} \\ \beta = \frac{\gamma(1 - \rho^2)(1 - \alpha)^2}{\gamma(1 - \rho^2) + (1 - \rho)^2} \end{array} \right.$$

- Now we have to derive the **innovation filter** of the observed process through the spectral factorization of its complex PSD:

$$S_X(z) = \sigma_w^2 \cdot \frac{\gamma(1 - \rho^2)}{\beta} \cdot \frac{(1 - \alpha z^{-1})(1 - \alpha z)}{(1 - \rho z^{-1})(1 - \rho z)} = \sigma_I^2 L_X(z) L_X(1/z)$$

■ We have immediately that:

$$\left\{ \begin{array}{l} \sigma_I^2 = \sigma_W^2 \cdot \frac{\gamma(1-\rho^2)}{\beta} = \sigma_W^2 \cdot \frac{\gamma(1-\rho^2) + (1-\rho)^2}{(1-\alpha)^2} \\ L_X(z) = \frac{1-\alpha z^{-1}}{1-\rho z^{-1}} \quad [\text{singularities within the unit circle}] \end{array} \right.$$

■ Then:

$$\begin{aligned} \frac{S_{XS}(z)}{L_X(1/z)} &= \frac{S_S(z)}{L_X(1/z)} = \frac{\sigma_S^2(1-\rho^2)}{(1-\rho z^{-1})(1-\rho z)} \cdot \frac{1-\rho z}{1-\alpha z} \\ &= \frac{\sigma_S^2(1-\rho^2)}{(1-\rho z^{-1})(1-\alpha z)} \end{aligned}$$

$$\left[\frac{S_{XS}(z)}{L_X(1/z)} \right]_+ = \left[\frac{\sigma_s^2(1-\rho^2)}{(1-\rho z^{-1})(1-\alpha z)} \right]_+ = \sigma_s^2(1-\rho^2) \left[\frac{1}{(1-\rho z^{-1})(1-\alpha z)} \right]_+$$

- To isolate the contribution that generates the causal part of the sequence, we have first to express it in terms of z^{-1} and then use the partial fraction expansion:

$$\frac{1}{(1-\rho z^{-1})(1-\alpha z)} = \frac{-(1/\alpha)z^{-1}}{(1-\rho z^{-1})\left(1-\frac{1}{\alpha}z^{-1}\right)} = \frac{c_1}{1-\rho z^{-1}} + \frac{c_2}{1-\frac{1}{\alpha}z^{-1}}$$

$$c_1 = \left[\left(1-\rho z^{-1}\right) \cdot \frac{1}{(1-\rho z^{-1})(1-\alpha z)} \right]_{z=\rho} = \frac{1}{1-\alpha\rho}$$

- Only the term with the pole ρ within the unit circle contributes to the causal part of the sequence, while the term with the pole in $1/\alpha$ contributes only to the non causal part. As a consequence, only the constant term c_1 needs to be evaluated.

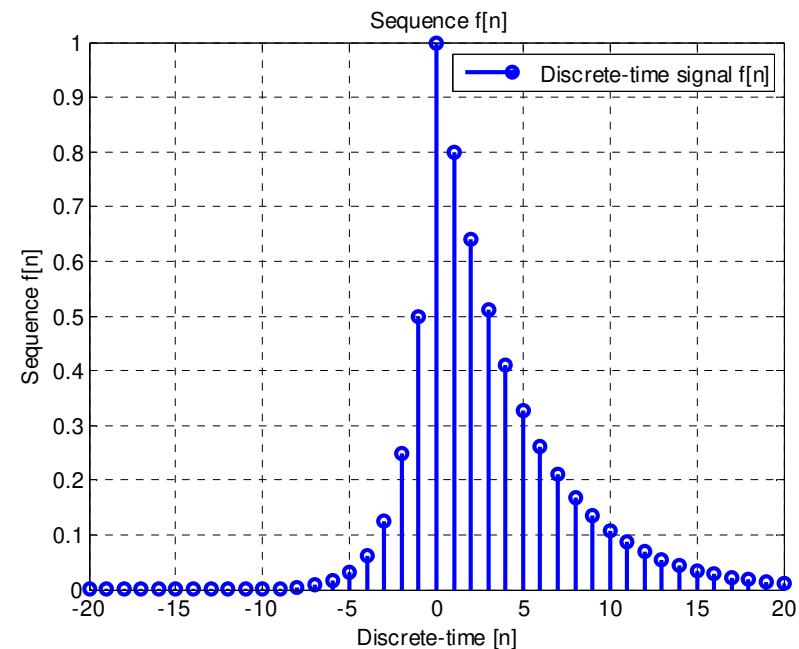
$$\begin{aligned}\left[\frac{S_{XS}(z)}{L_X(1/z)} \right]_+ &= \sigma_s^2(1 - \rho^2) \left[\frac{1}{(1 - \rho z^{-1})(1 - \alpha z)} \right]_+ \\ &= \sigma_s^2(1 - \rho^2) \left[\frac{c_1}{1 - \rho z^{-1}} + \frac{c_2}{1 - \frac{1}{\alpha} z^{-1}} \right]_+ \\ &= \sigma_s^2(1 - \rho^2) \frac{c_1}{1 - \rho z^{-1}} = \frac{\sigma_s^2(1 - \rho^2)}{1 - \alpha \rho} \cdot \frac{1}{1 - \rho z^{-1}}\end{aligned}$$

- In order to better understand this result, suppose to have the following non causal sequence $f[n]$, i.e. $f[n]$ is composed of a causal part and of a anti-causal part:

$$f[n] = f_{ac}[n] + f_c[n] = \begin{cases} c_2 \left(\frac{1}{\alpha} \right)^n, & n < 0 \\ c_1 \rho^n, & n \geq 0 \end{cases}$$

with $|\alpha| < 1$ and $|\rho| < 1$

$$c_1 = 1, c_2 = 1$$
$$\alpha = 0.5, \rho = 0.8$$



- The bilateral Z transform of $f[n]$ is:

$$\begin{aligned} F(z) &= \sum_{n=-\infty}^{+\infty} f[n]z^{-n} = c_2 \sum_{n=-\infty}^{-1} \left(\frac{1}{\alpha}\right)^n z^{-n} + c_1 \sum_{n=0}^{+\infty} \rho^n z^{-n} \\ &= c_2 \sum_{k=1}^{+\infty} \left(\frac{1}{\alpha}\right)^{-k} z^k + c_1 \sum_{n=0}^{+\infty} \rho^n z^{-n} \\ &= c_2 \sum_{k=0}^{+\infty} \left(\frac{1}{\alpha}\right)^{-k} z^k - c_2 + c_1 \sum_{n=0}^{+\infty} \rho^n z^{-n} \\ &= c_2 \sum_{k=0}^{+\infty} \alpha^k z^k - c_2 + c_1 \sum_{n=0}^{+\infty} \rho^n z^{-n} \\ &= c_2 \sum_{k=0}^{+\infty} (\alpha z)^k - c_2 + c_1 \sum_{n=0}^{+\infty} (\rho z^{-1})^n \end{aligned}$$

$$\begin{aligned} F(z) &= \sum_{n=-\infty}^{+\infty} f[n]z^{-n} \\ &= c_2 \sum_{k=0}^{+\infty} (\alpha z)^k - c_2 + c_1 \sum_{n=0}^{+\infty} (\rho z^{-1})^n = \frac{c_2}{1 - \alpha z} - c_2 + \frac{c_1}{1 - \rho z^{-1}} \end{aligned}$$

- The first series converges if $|\alpha z| < 1$, the second series converges if $|\rho z^{-1}| < 1$, that is the ROC of the bilateral Z transform of the sequence $f[n]$ is given by the portion of the complex plane in which both the two conditions are verified:

$$|\alpha z| < 1 \Rightarrow |z| < \frac{1}{|\alpha|}, \quad |\rho z^{-1}| < 1 \Rightarrow |\rho| < |z|$$

$$ROC: |\rho| < |z| < \frac{1}{|\alpha|} \quad [\text{annular region that includes the unit circle}]$$



$$\begin{aligned} F(z) &= \frac{c_2}{1 - \alpha z} - c_2 + \frac{c_1}{1 - \rho z^{-1}} \\ &= \frac{c_2 \alpha z}{1 - \alpha z} + \frac{c_1}{1 - \rho z^{-1}} \\ &= -\frac{c_2}{1 - \frac{1}{\alpha} z^{-1}} + \frac{c_1}{1 - \rho z^{-1}}, \quad |\rho| < |z| < \frac{1}{|\alpha|} \end{aligned}$$

- Consequently, the anti-causal part of the sequence contributes with a term that has a pole outside the unit circle ($1/\alpha$), while the causal part of the sequence contributes with a term that has a pole within the unit circle (ρ).

$$f[n] = f_{ac}[n] + f_c[n] = \begin{cases} c_2 \left(\frac{1}{\alpha} \right)^n, & n < 0 \\ c_1 \rho^n, & n \geq 0 \end{cases}$$

$$f_{ac}[n] = c_2 \left(\frac{1}{\alpha} \right)^n u[-n-1], \quad f_c[n] = c_1 \rho^n u[n]$$

$$[F(z)]_+ = F_c(z) = ZT \{f_c[n]\}$$

$$F_c(z) = [F(z)]_+ = \left[\frac{c_2 \alpha z}{1 - \alpha z} + \frac{c_1}{1 - \rho z^{-1}} \right]_+ = \frac{c_1}{1 - \rho z^{-1}} \stackrel{ZT}{\Leftrightarrow} f_c[n] = c_1 \rho^n u[n]$$

- The system function $H(z)$ of the causal linear IIR Wiener filter for an AR(1) process observed in the presence of additive noise is an IIR(1) filter given by:

$$\begin{aligned} H(z) &= \frac{1}{\sigma_I^2 L_X(z)} \left[\frac{S_{XS}(z)}{L_X(1/z)} \right]_+ = \frac{1}{\sigma_I^2 L_X(z)} \left[\frac{S_S(z)}{L_X(1/z)} \right]_+ \\ &= \frac{\gamma(1-\rho^2)(1-\alpha)^2}{(1-\alpha\rho) \left[\gamma(1-\rho^2) + (1-\rho)^2 \right]} \cdot \frac{1}{1-\alpha z^{-1}} = \frac{c_0}{1-\alpha z^{-1}} \end{aligned}$$

$$h[n] = c_0 \alpha^n u[n] \quad \xleftrightarrow{ZT} \quad H(z) = \frac{c_0}{1-\alpha z^{-1}}$$

$$h[n] = c_0 \alpha^n u[n] \quad \xleftrightarrow{ZT} \quad H(z) = \frac{c_0}{1 - \alpha z^{-1}}$$

$$\hat{S}[n] = \sum_{k=0}^{+\infty} h[k] X[n-k] = c_0 \sum_{k=0}^{+\infty} \alpha^k X[n-k]$$

- The output of the causal IIR Wiener filter can be rewritten in a recursive form as:

$$\hat{S}(z) = H(z)X(z) = \frac{c_0 X(z)}{1 - \alpha z^{-1}} \quad \rightarrow \quad \hat{S}(z)(1 - \alpha z^{-1}) = c_0 X(z)$$

$$\rightarrow \hat{S}(z) = \alpha z^{-1} \hat{S}(z) + c_0 X(z) \quad \rightarrow \quad \hat{S}[n] = \alpha \hat{S}[n-1] + c_0 X[n]$$

- The MSE of the estimate of the causal IIR Wiener filter for an AR(1) process is given by:

$$\sigma_{\varepsilon}^2 = R_S[0] - \sum_{k=0}^{+\infty} h[k] R_S[k] = \sigma_S^2 \left(1 - c_0 \sum_{k=0}^{+\infty} \alpha^k \rho^k \right) = \sigma_S^2 \left(1 - \frac{c_0}{1 - \alpha\rho} \right)$$

- In the particular case: $\sigma_S^2 = 2$, $\rho = 0.8$, $\sigma_W^2 = 2 \Rightarrow \gamma = 1$

$$\begin{cases} \sigma_I^2 = \sigma_W^2 \cdot \frac{\gamma(1-\rho^2)}{\beta} = 3.2 \\ c_0 = \frac{\gamma(1-\rho^2)(1-\alpha)^2}{(1-\alpha\rho)[\gamma(1-\rho^2) + (1-\rho)^2]} = 0.375 \end{cases} \Rightarrow \alpha = 0.5, \beta = 0.225$$


- The system function $H(z)$ of the causal optimal IIR linear filter and the impulse response $h[n]$ are given by:

$$h[n] = 0.375 \cdot (0.5)^n u[n] \quad \xleftrightarrow{ZT} \quad H(z) = \frac{0.375}{1 - 0.5z^{-1}}$$

- The MSE is given by:

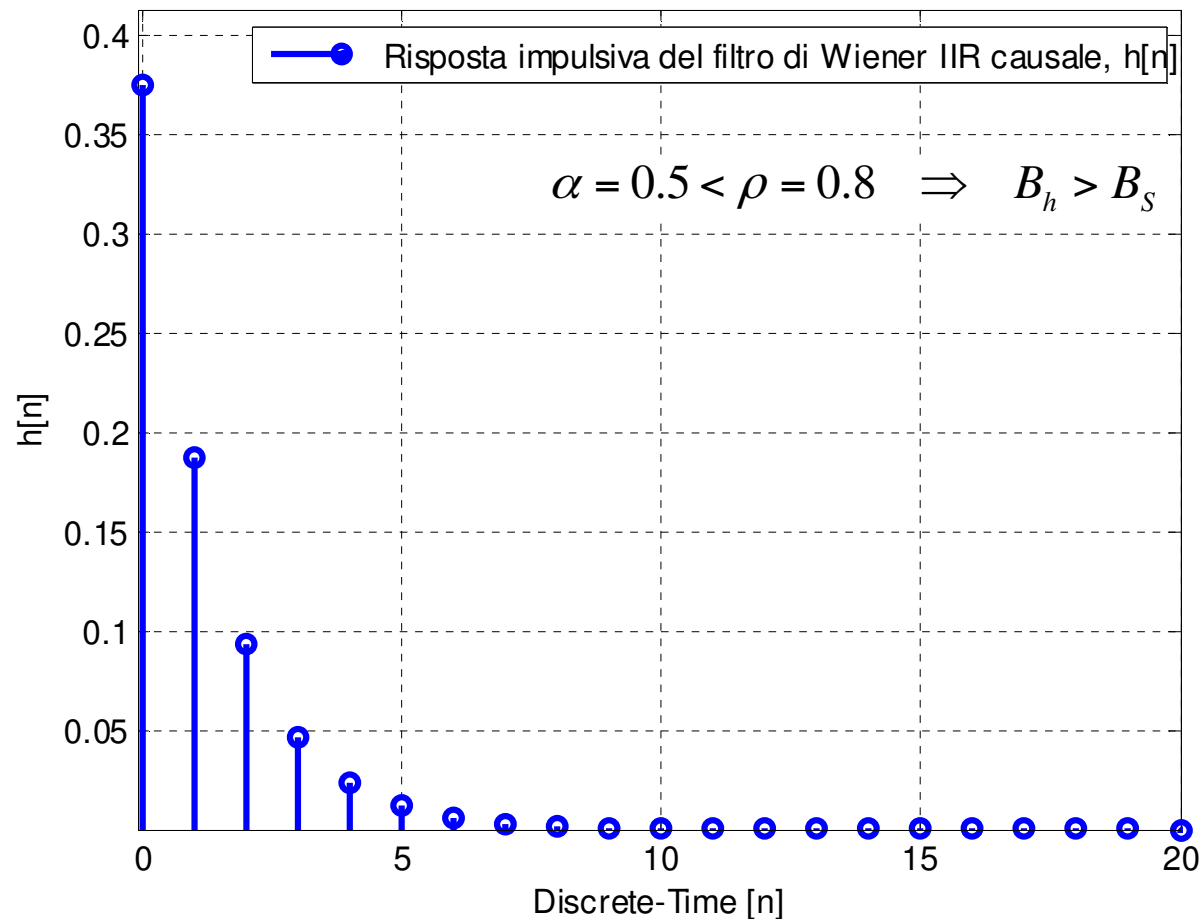
$$\sigma_\varepsilon^2 = \sigma_s^2 \left(1 - \frac{c_0}{1 - \alpha\rho} \right) = 0.75$$

- It can be noted that: *sono intermedie*

$$\underbrace{\sigma_{\varepsilon, FIR(2)}^2 = 0.7647}_{\downarrow \text{Fatto}} > \underbrace{\sigma_{\varepsilon, IIRca}^2 = 0.75}_{\downarrow \text{Non usa pass-fut.}} > \underbrace{\sigma_{\varepsilon, IIRnc}^2 = 0.6}_{\downarrow \text{Fatto}}$$

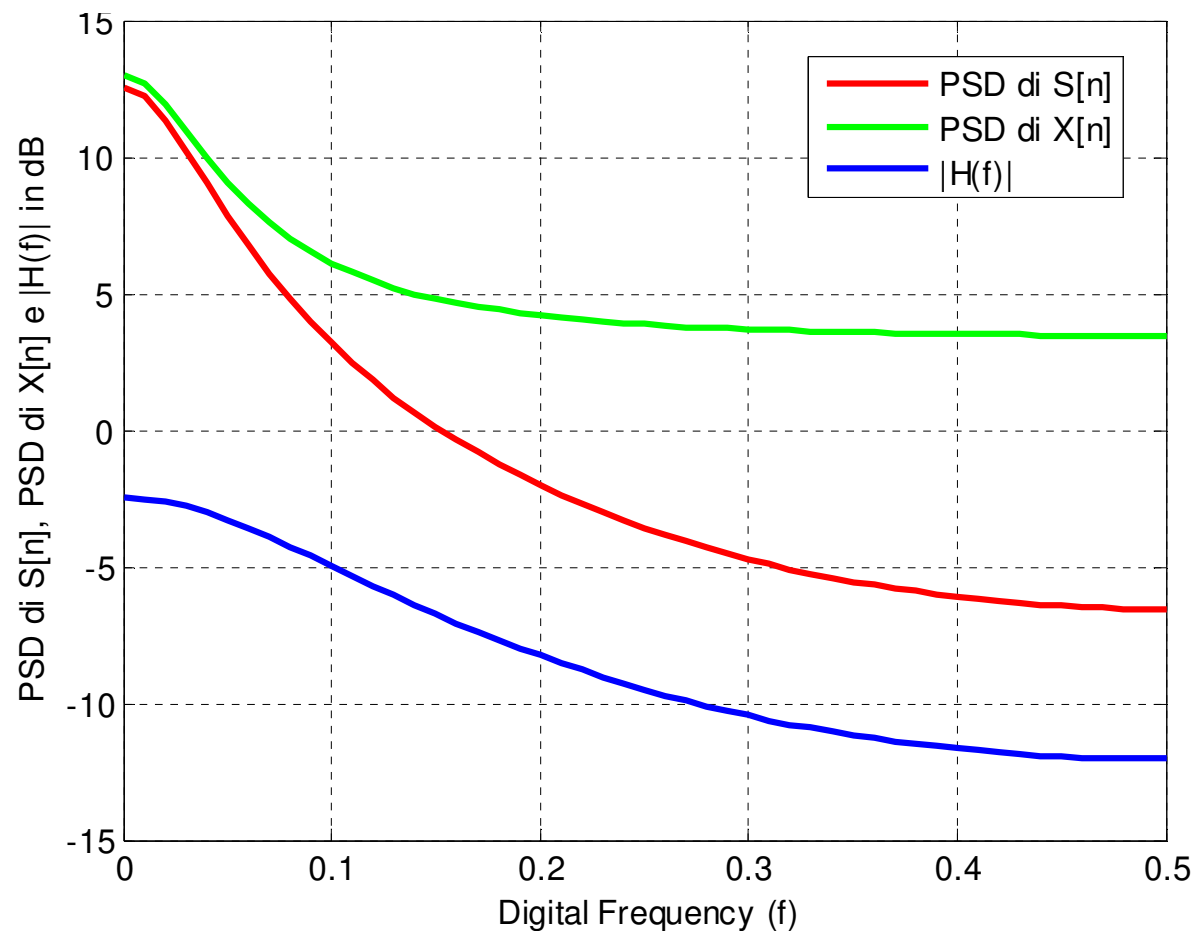
$$\sigma_s^2 = 2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 1 \Rightarrow \gamma_{dB} = 0 \text{ dB}$$

Impulse Response (IR) of the causal IIR Wiener filter



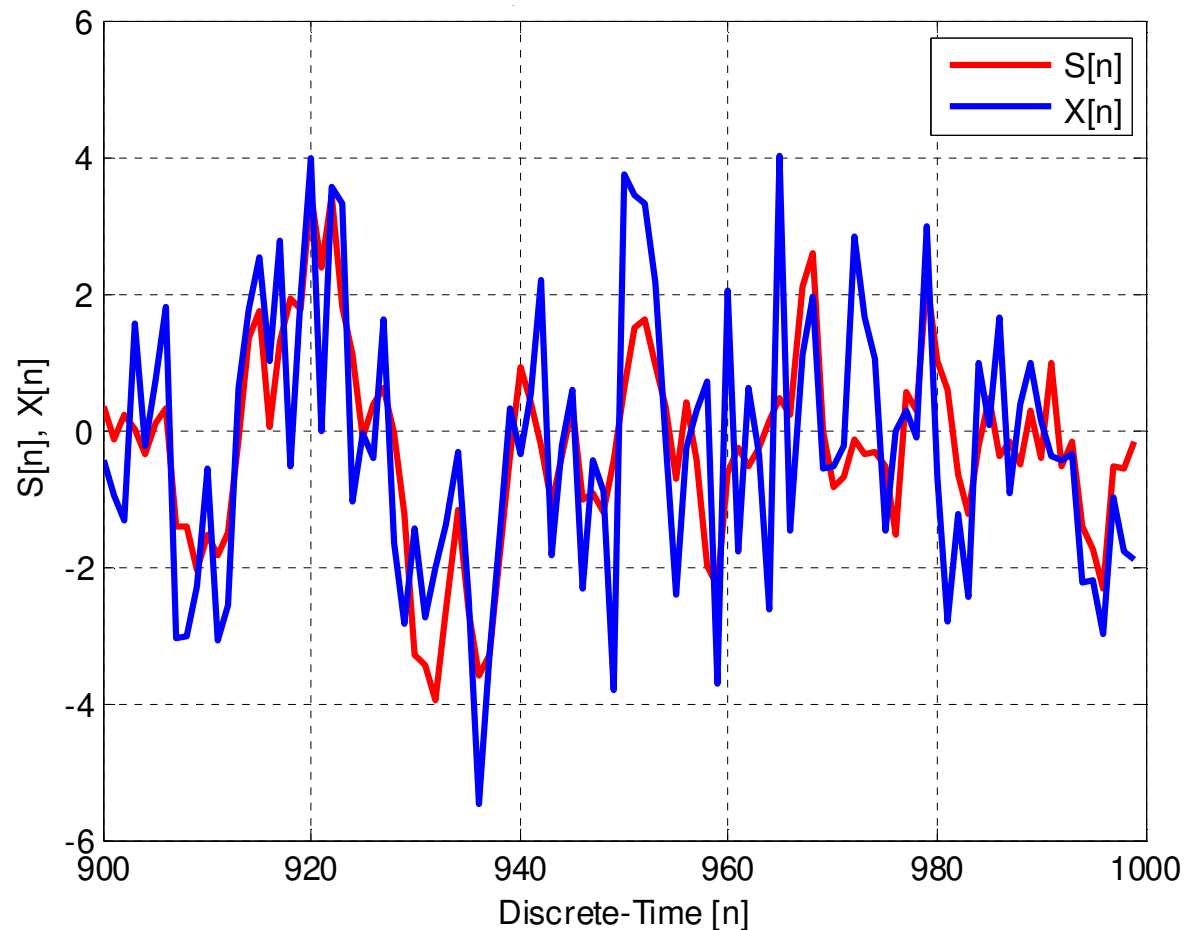
$$\sigma_s^2 = 2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 1 \Rightarrow \gamma_{dB} = 0 \text{ dB}$$

PSD of $S[n]$, PSD of $X[n]$, and $|H(f)|$ of the IIR causal Wiener filter



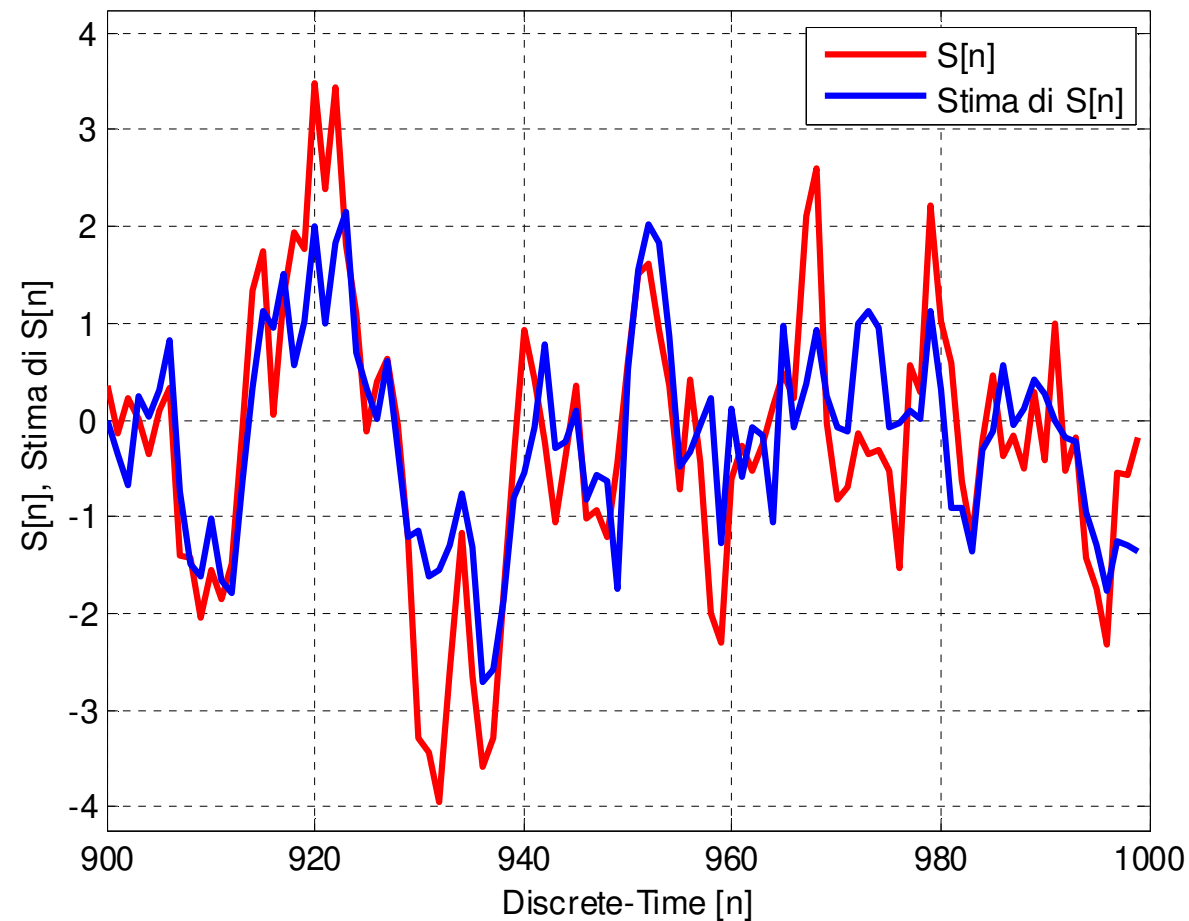
$$\sigma_s^2 = 2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 1 \Rightarrow \gamma_{dB} = 0 \text{ dB}$$

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



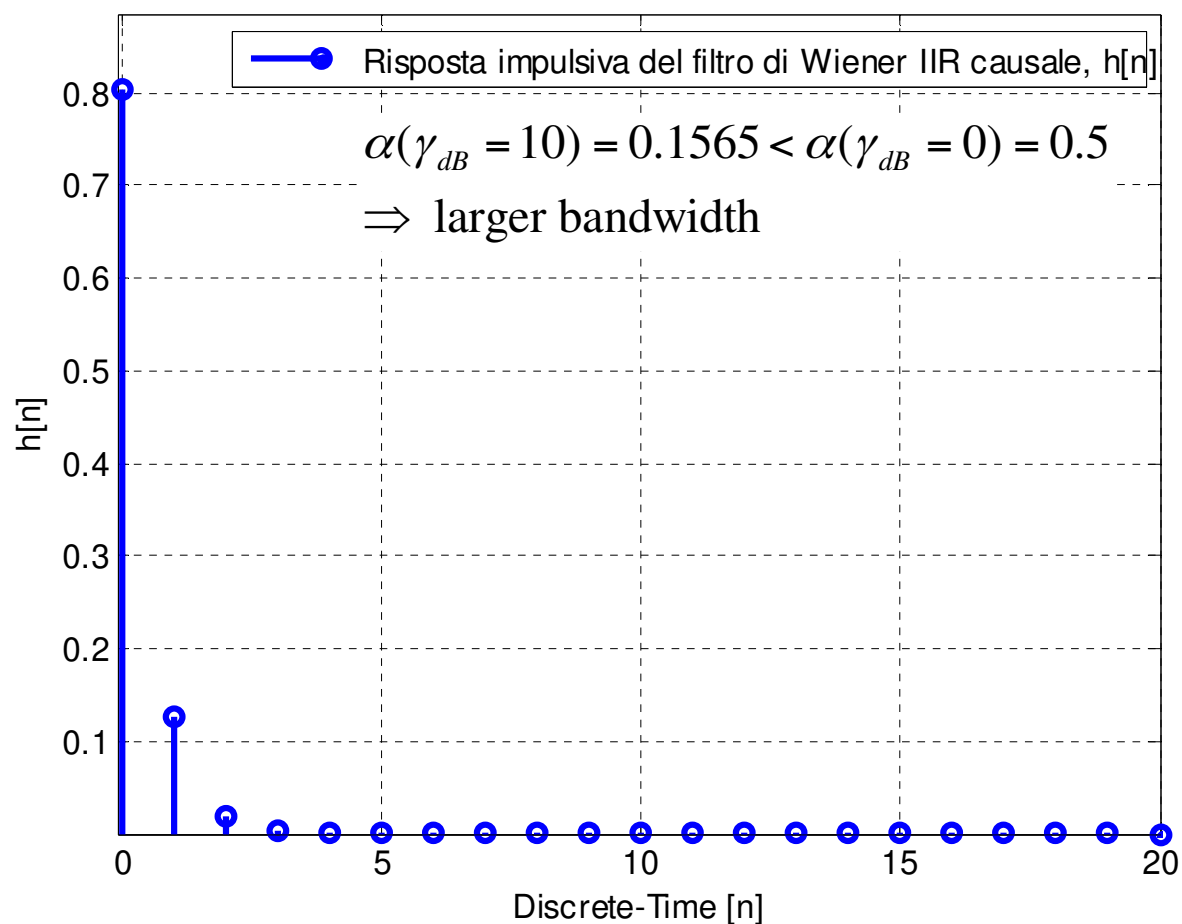
$$\sigma_s^2 = 2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 1 \Rightarrow \gamma_{dB} = 0 \text{ dB}$$

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



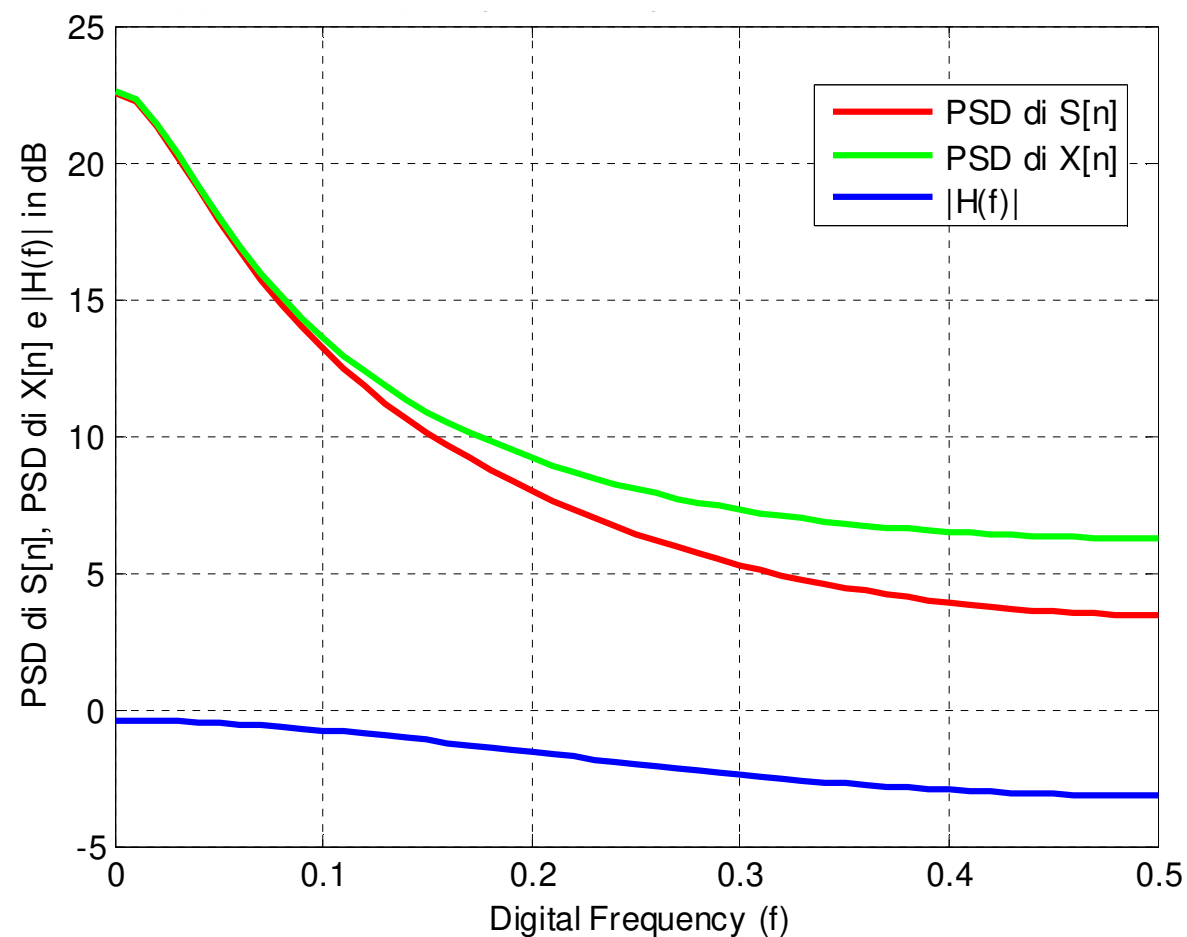
$$\sigma_s^2 = 20, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 10 \Rightarrow \gamma_{dB} = 10 \text{ dB}$$

Impulse Response (IR) of the causal IIR Wiener filter



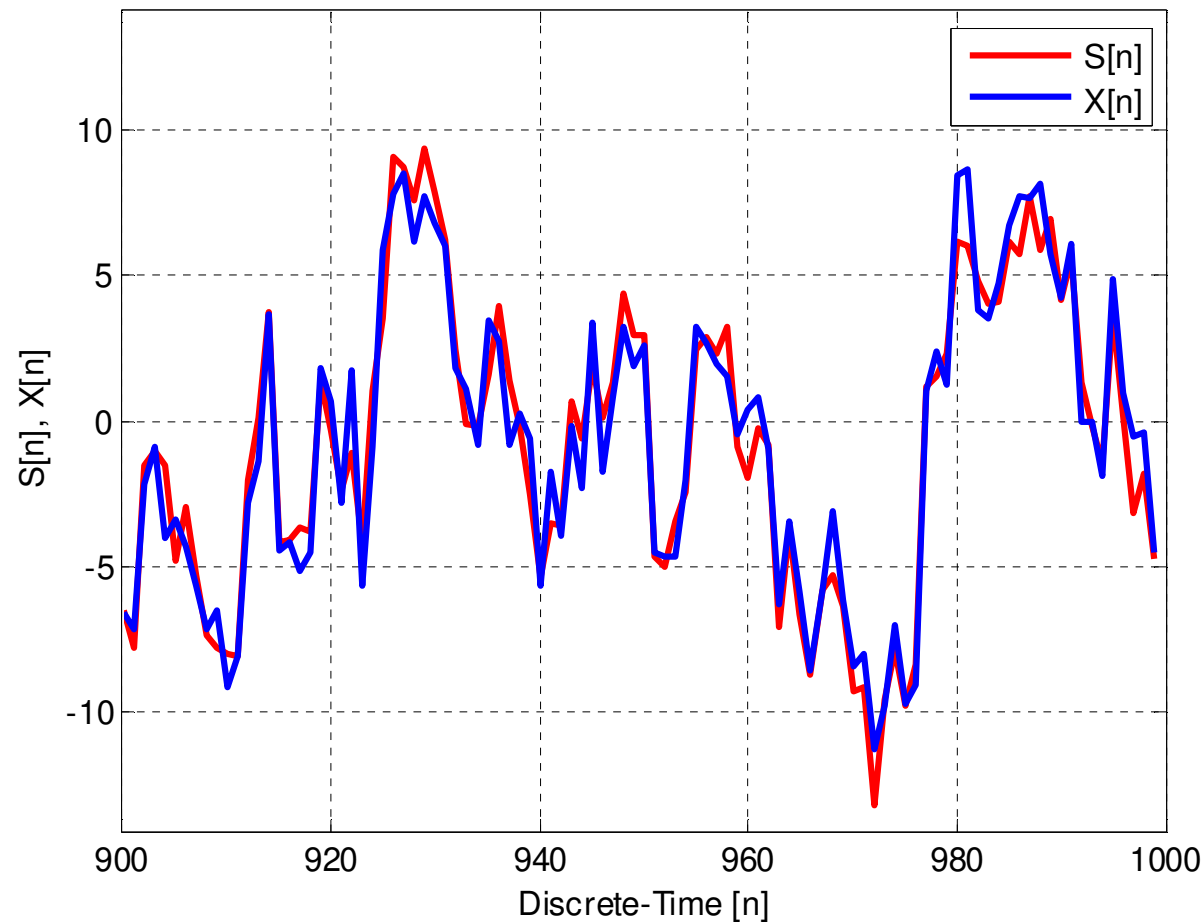
$$\sigma_s^2 = 20, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 10 \Rightarrow \gamma_{dB} = 10 \text{ dB}$$

PSD of $S[n]$, PSD of $X[n]$, and $|H(f)|$ of the IIR causal Wiener filter



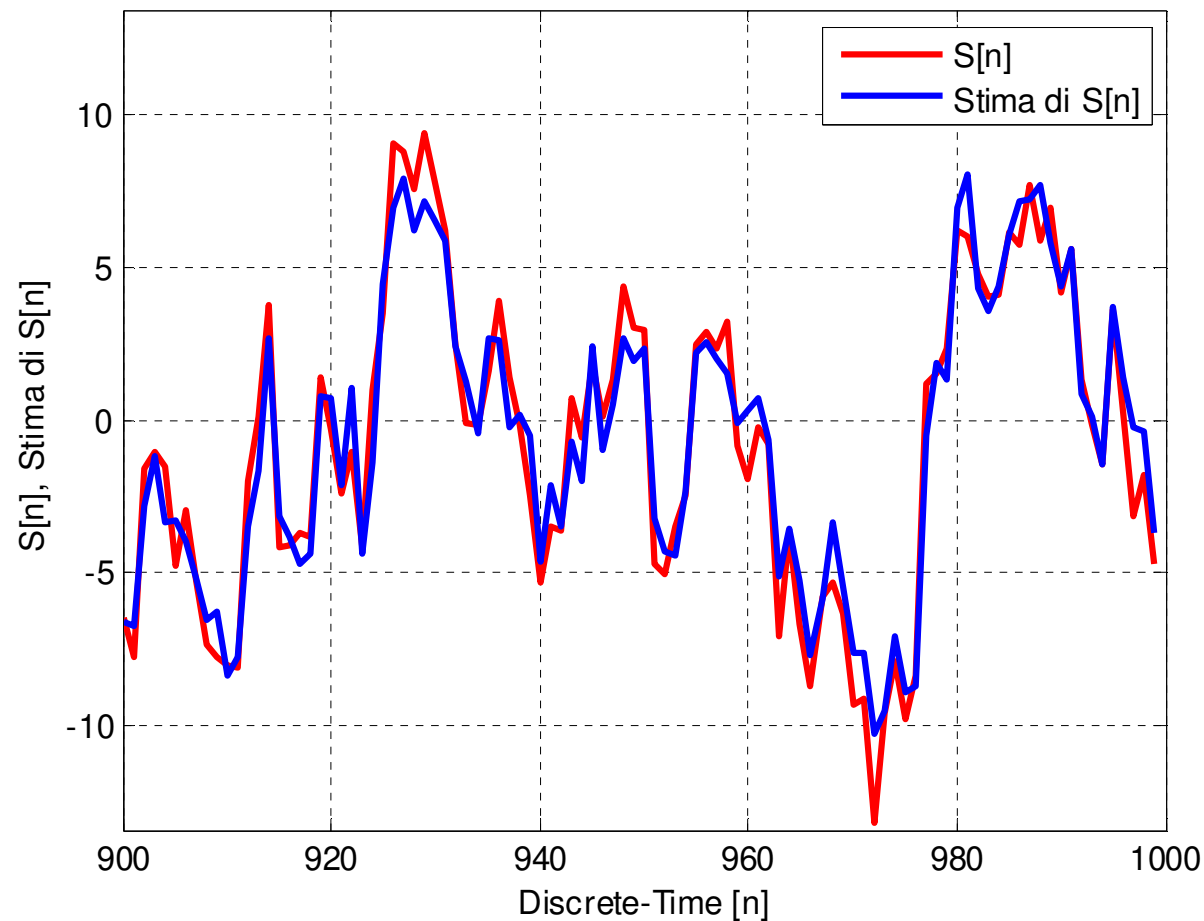
$$\sigma_s^2 = 20, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 10 \Rightarrow \gamma_{dB} = 10 \text{ dB}$$

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



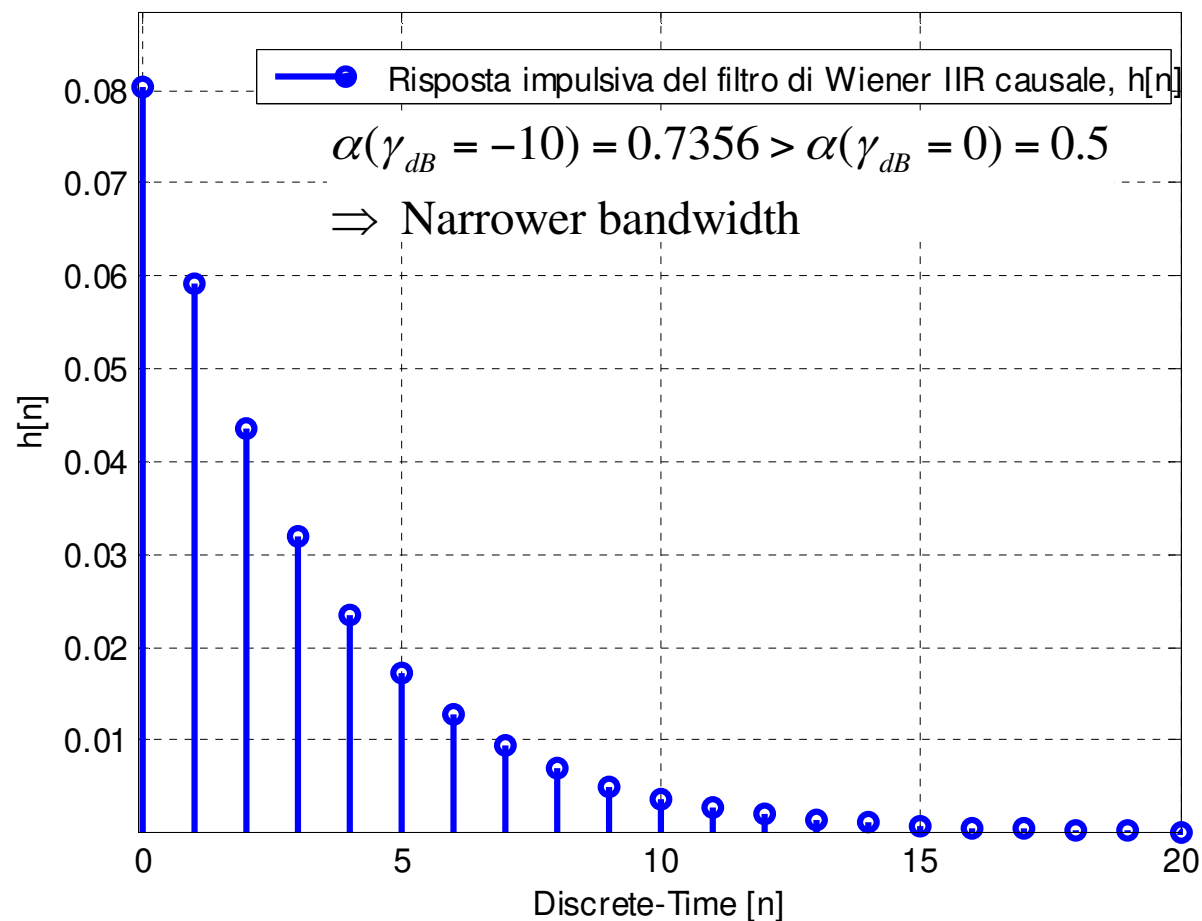
$$\sigma_s^2 = 20, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 10 \Rightarrow \gamma_{dB} = 10 \text{ dB}$$

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



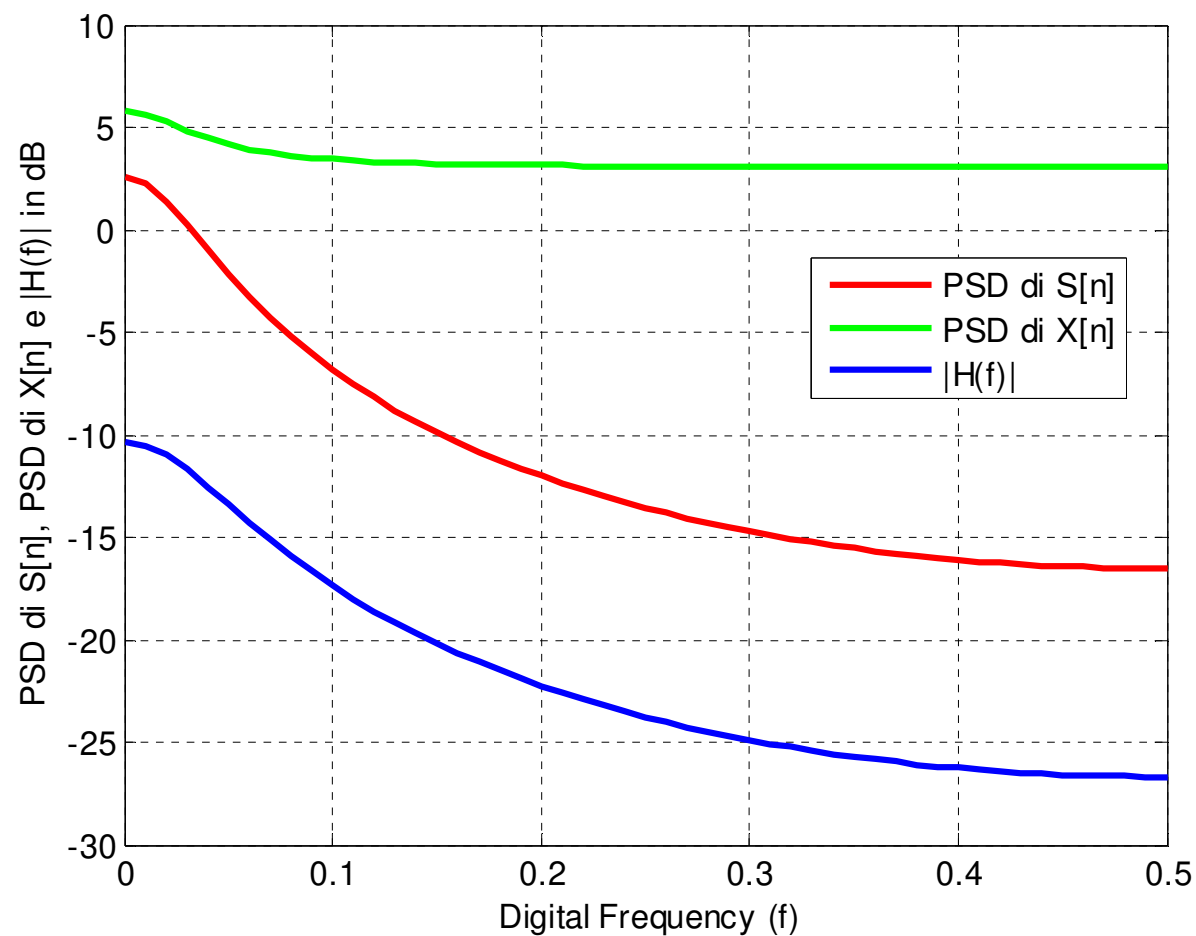
$$\sigma_s^2 = 0.2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 0.1 \Rightarrow \gamma_{dB} = -10 \text{ dB}$$

Impulse Response (IR) of the causal IIR Wiener filter



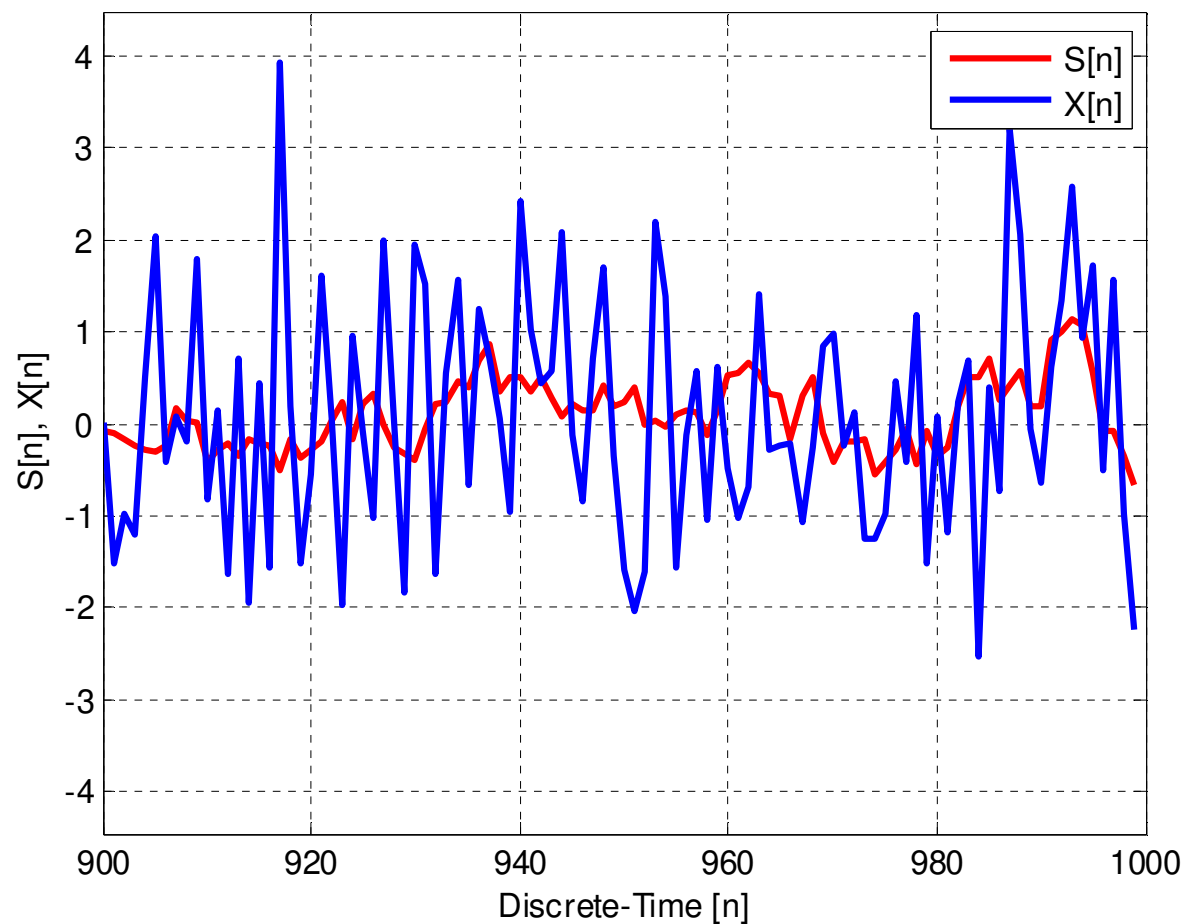
$$\sigma_s^2 = 0.2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \Rightarrow \gamma = 0.1 \Rightarrow \gamma_{dB} = -10 \text{ dB}$$

PSD of $S[n]$, PSD of $X[n]$, and $|H(f)|$ of the IIR causal Wiener filter



$$\sigma_s^2 = 0.2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \quad \Rightarrow \quad \gamma = 0.1 \quad \Rightarrow \quad \gamma_{dB} = -10 \text{ dB}$$

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN



$$\sigma_s^2 = 0.2, \quad \rho = 0.8, \quad \sigma_w^2 = 2 \quad \Rightarrow \quad \gamma = 0.1 \quad \Rightarrow \quad \gamma_{dB} = -10 \text{ dB}$$

$X[n] = S[n] + W[n]$, $S[n]$ = Gaussian AR(1) process, $W[n]$ = AWGN

